FAMILY HANDBOOK

Wiggins Survival Book

Staying warm, fed, and safe on the eastern plains of Colorado
For our family — Morgan County, Colorado

Edition: October 2026

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General guidance for one family, not a substitute for a doctor, licensed electrician, certified chimney sweep, or official emergency instructions. Call 911 in an emergency.

CHAPTER 1

How to Use This Book & the First 72 Hours

Who this book is for

This is the survival handbook for **our family in Wiggins, Colorado** (Morgan County, eastern plains, about 4,500 feet). It is written for:

- **Daniel** and his **wife**
- **The kids:** a newborn (breastfeeding, born late September 2026), a 1-year-old (on formula), a 2-year-old (on regular milk), and a 6-year-old
- **Grandma and Grandpa**, next door

We have **chickens** (about 6–12 eggs a day), **many rabbits**, a **propane tank** feeding a modern gas furnace and central AC, and an **old wood stove** in the basement that is not hooked up yet but **we do have a chimney**.

Our biggest local threats are **blizzards and deep cold**, **long power outages**, **high wind**, **tornadoes and hail**, **grass fires**, **summer heat and drought**, and **flooding along the South Platte**.

How to read it

- Read it **once, all the way through, now** while times are normal. Most of what keeps a family alive happens **before** an emergency.
- Each chapter ends with a **"Do it now" checklist**. Check things off in pencil.
- **The Home Worksheet** (in the phone app's Appendix — it is not in the printed book) is where we write down our shutoffs, propane, well, chimney, medicines, and contacts. Fill it in and keep it updated.
- Keep **one printed copy** in the kitchen and **one at Grandma and Grandpa's**. Put it in a gallon zip bag so it survives water.
- Phones die. Paper doesn't. If the power is out, use the paper copy.

> **Rule of thumb:** This book is a guide, not a doctor, electrician, or chimney sweep. **Use professionals now, while they are available.** If services are running and you're worried about a baby, call the doctor or 911. If they aren't, Chapters 8 and 9 are written for that.

The rule of threes (what kills first)

People usually die in this order when things go bad:

1. **Air / carbon monoxide** — minutes. *(Generators, grills, or heaters used indoors.)*

2. **Cold (exposure)** — hours in a Colorado blizzard. Babies and older adults lose heat fastest.
3. **Water** — about 3 days. Babies dehydrate much faster.
4. **Food** — weeks.

So the priority list is always: **safe air → warmth → water → food → everything else.**

Survival mindset (from the U.S. Army Survival Manual)

The Army's survival manual (FM 21-76, now ATP 3-50.21) builds everything on the word **SURVIVAL**:

- **S**ize up the situation, yourself, and your gear.
- **U**se all your senses — and **U**ndue haste makes waste. Don't rush into mistakes.
- **R**emember where you are (and where help, water, and fuel are).
- **V**anquish fear and panic.
- **I**mprovise.
- **V**alue living — stubbornness to get the family through matters.
- **A**ct like the natives — learn how locals (old ranchers, Grandma and Grandpa) handled it.
- **L**ive by your wits, but for now, **learn basic skills.**

When something goes wrong, use **STOP: Stop, Think, Observe, Plan** — before acting.

Lessons from people who lived it

- **Texas, February 2021 (Winter Storm Uri):** millions lost power for days in deep cold. Families learned that **no power means no heat and often no water**, pipes burst, and **diesel fuel can gel in very cold weather** (keep anti-gel additive in diesel trucks and generators). Many died of **hypothermia and carbon monoxide** from improvised heating. *(The Prepared; CDC)*
- **The 1888 "Schoolhouse Blizzard" on the Great Plains** hit on a mild day with almost no warning; many people, including children walking home from school, died caught outside. **Lesson: when a plains blizzard hits, stay put in shelter.**
- **Eastern Colorado, December 2006:** two blizzards struck about a week apart. See Chapter 12.

The first hour

1. **Get everyone together** (both houses). Count heads: 8 people.
2. **Check for danger:** gas smell, fire, injuries, downed power lines. If you smell gas: **get out, don't flip switches, call 911 / the propane company from outside.**
3. **Turn on the battery NOAA Weather Radio** (Chapter 10) to learn what is happening.
4. **Fill water containers and bathtubs now** while water pressure still works (Chapter 2).

5. **Decide: stay or go?** Most of the time we **shelter in place**. Leave if: fire is approaching, floodwater is rising, the house is unsafe, or officials order evacuation.

The first 24 hours

- **Pick one warm room** if heat is out (Chapter 3). Close off the rest of the house.
- **Keep the fridge and freezer shut.** A full freezer holds food safely about 48 hours, a half-full one about 24 hours, if the door stays closed. A fridge holds about 4 hours. *(USDA)*
- **Check on babies every hour:** warm? feeding? wet diapers?
- **Ration phone battery.** Text instead of calling. Turn on low-power mode.
- **Check animals:** water for chickens and rabbits. In winter, it freezes fast.
- **Write down what happened and when** on the log page at the back.

The first 72 hours

- **Make a supply count** (Chapter 13 inventory sheets): water, formula, diapers, food, propane %, firewood, medicines.
- **Set a routine:** meals, baby feeds, chores, sleep shifts, someone on watch for fire/CO.
- **Combine households if heat or water is short.** Two houses heated is twice the fuel. Usually it is better for everyone to live in **one** house.
- **Start conserving immediately.** It is easier to ease up later than to run out.

If we must leave (evacuation)

Keep a **go-bag** for each person, ready by the door (see Chapter 13). Pick meeting places:

- **Spot 1:** End of our driveway (fire in the house).
- **Spot 2:** A place in town (fill in on the worksheet).
- **Spot 3:** Out of the area — a relative or friend (fill in).

Leave a note on the door saying where we went and when.

Do it now

- ☐ Read this whole book once.
- ☐ Fill in the Home Worksheet.
- ☐ Print two copies; one at each house.
- ☐ Buy a battery/hand-crank **NOAA Weather Radio**.
- ☐ Put **battery CO alarms** on every level of both houses.
- ☐ Sign up for **Morgan County emergency alerts** (ask the County Sheriff's Office / Office of Emergency Management) and turn on Wireless Emergency Alerts on all phones.

CHAPTER 2

Water

You can live weeks without food but only **about 3 days without water** — and babies, nursing moms, and older adults much less. Water is the first thing to secure.

How much water we need

The standard rule (FEMA / Ready.gov): **at least 1 gallon per person per day** for drinking and basic hygiene. Store **at least a 2-week supply** if you can.

Nursing mothers, sick people, and anyone in hot weather need **more**. Plan extra.

Who	Gallons/day (minimum)	2 weeks
7 people eating/drinking (all but newborn)	7	98
Extra for nursing mom, formula, cooking, heat	+2	+28
Family total	**~9**	**~126 gallons**
Chickens & rabbits	see Chapters 5–6	

Our goal: about 125+ gallons stored for people, plus water for animals.

Storing water

- Use **food-grade** containers: store-bought bottled water, 5–7 gallon water jugs, or 55-gallon food-grade barrels (blue).
- **Do not use** milk jugs (they break down and leak) or containers that held chemicals.
- Store in a cool, dark place, off concrete (put on a pallet or wood).
- **Replace stored tap water every 6 months.** Unopened commercial bottled water: follow the date. *(Ready.gov)*
- **Don't let it freeze and split** — keep inside the heated part of the house in winter, or leave room for expansion.
- Hidden water in the house: the **water heater tank** (often 40–50 gallons), pipes, and toilet tanks (**not** bowls; and not if you use tank cleaners).

Draining the water heater in an emergency

1. **Turn off the gas/power to the water heater first.**
2. Close the cold-water inlet valve on top.
3. Open a hot faucet upstairs to let air in.
4. Drain from the valve at the bottom into a clean container. The first water may have sediment — let it settle.

Making water safe

If the water might be contaminated (well flooded, town "boil water" notice, river or stock tank water):

1. Boil (best)

- Let cloudy water settle, then filter it through a clean cloth or coffee filter.
- Bring to a **rolling boil for 1 minute**. *(Above 6,500 ft boil 3 minutes; Wiggins is below that.) (CDC)*
- Let cool. Pour between clean containers to improve taste.

2. Bleach (when you can't boil)

Use **plain, unscented household bleach** (no "splash-less" or scented). Check the label for the percent.

Bleach strength	Per 1 gallon of clear water	Per 5 gallons
6%	**8 drops** (about 1/8 tsp)	40 drops (½ tsp)
8.25%	**6 drops**	30 drops (about ⅓ tsp)

- **Double the dose if the water is cloudy or very cold.**
- Stir and **wait 30 minutes.** It should smell slightly of chlorine. If not, repeat and wait another 15 minutes. *(EPA)*
- Bleach loses strength over time — replace yearly.

3. Filters

A good gravity or squeeze filter (such as those rated to remove bacteria and protozoa) is very useful, but **most filters do NOT remove viruses or chemicals.** For the safest water: **filter, then boil or bleach.**

What none of these fix

Chemicals, fuel, fertilizer, or pesticides (common after floods in farm country) are **not** removed by boiling or bleach. If water smells of fuel or chemicals, **don't use it.**

Water for the babies

- **The newborn needs no water** — only breast milk. Never give a young baby plain water.
- **Formula water:** use **bottled water** or water that has been **boiled**. If unsure of water safety, use **ready-to-feed formula** (needs no water at all) — this is the best emergency choice. See Chapter 4.
- Wash bottles in safe water; boil bottles and nipples if you can't wash properly.

If we have a private well

Most wells use an **electric submersible pump. No power = no water.** Plan one of these **now**:

1. **Generator + transfer switch or interlock** installed by a licensed electrician so the well pump can run. Well pumps often need a large generator — check the pump's label (volts/horsepower) and ask the electrician.
2. **Manual well hand pump** installed alongside the electric pump. Several brands make pumps that share the well casing. **Whether this works depends on our static water level** — deep water levels need special (and more expensive) pumps. Get our well log (worksheet) and ask a licensed well contractor.
3. **Solar well pump** (costly, but works for years).
4. **Stored water** (above) to bridge the gap.

After a flood: a flooded well may be contaminated. Don't drink it until it is disinfected (shock chlorinated) and tested. Colorado labs and CSU Extension give guidance; the county health department can help.

Freezing: keep the well house/pressure tank area above freezing — insulate, use a heat lamp **only if** fire-safe, and keep the pump area dry.

If we have town water

- Town water usually keeps flowing for a while in outages, but pressure may drop and **boil-water notices** may come. Listen to the radio.
- **Fill everything when trouble starts.**
- Know where the main shutoff is — if a pipe breaks, shut it off fast.

Winter: keeping pipes from freezing

- Keep the house above freezing. Open cabinet doors under sinks on outside walls.
- Let a faucet **drip** on very cold nights (if water supply allows).
- **If we lose heat for days:** shut off the main water, open all faucets (top floor first, then lowest), and flush toilets to drain lines. Put a little RV antifreeze in drains and toilet bowls if you have it. *(Do not drink RV antifreeze.)*
- **Snow is water:** melt it — but treat/boil it like any untreated water, and it takes a lot of snow (and fuel) to make a little water. Never eat snow to stay hydrated — it chills the body.

Sanitation (just as important as water)

Dirty hands and human waste spread diarrhea, which can kill babies quickly through dehydration.

- **Wash hands** with soap and water after the toilet, diapers, animals, and before food. Hand sanitizer (60%+ alcohol) if water is short — but it doesn't work well on dirty hands.
- **If toilets can't flush:** use a **bucket toilet**: a 5-gallon bucket lined with two heavy trash bags. Add kitty litter, sawdust, or wood ash after each use. Seal and store bags outside away from water and animals. Keep a lid on.
- **Don't** put human waste or used diapers in the chicken or garden compost.
- Keep the bucket toilet **away from the kitchen and water storage.**
- Dish water: wash dishes in hot soapy water, rinse in water with a little bleach (about 1 tablespoon bleach per gallon of water), air-dry.

River and stock water

The **South Platte** and ditches near farm fields can carry farm runoff and animal waste. Only use them if you must — **settle, filter, then boil.** Never let kids play in floodwater.

Do it now

- [ ] Store ~125 gallons of drinking water (start with cases of bottled water and a few 5–7 gal jugs).
- [ ] Buy unscented bleach and an eyedropper; tape the dose chart to the jug.
- [ ] Buy a water filter (bacteria + protozoa) and a large stockpot for boiling.
- [ ] Learn how to drain our water heater.
- [ ] **Well:** get the well log; talk to an electrician (generator inlet) and a well contractor (hand pump).
- [ ] Bucket toilet kit: 2 buckets with lids, heavy bags, kitty litter, TP, gloves.
- [ ] Mark the main water shutoff with a tag.

CHAPTER 3

Heat & Shelter in Winter

On the eastern plains, a winter storm with wind can drop wind chills far below zero. **Without heat, a house can become dangerously cold within a day or two.** Babies and Grandma and Grandpa are at highest risk. This chapter covers keeping warm with propane, with the wood stove, and with neither.

> ⚠ **Carbon monoxide (CO) is one of the biggest killers in winter power outages.** It has no smell. Every year families die from generators, grills, and heaters used indoors. **Put battery-powered CO alarms on every level, and near every sleeping area, in both houses — now.**

Our heat sources, in order

1. **Propane furnace** (normal) — needs **electricity** to run.
2. **Wood stove** (basement) — reconnected to our existing chimney, **after professional inspection**.
3. **Indoor-rated propane heater** (backup, used exactly per the label).
4. **Body heat, layers, and one insulated room** (always works).

1. The propane furnace

No power = no furnace

A modern gas furnace uses electricity for the **igniter, controls, and blower fan**. When the power is out, **it will not run** even with a full propane tank.

Fix (do this now, if we choose a generator): have a **licensed electrician** install a **generator inlet with a transfer switch or interlock**, so a generator outside can power the furnace (and well pump, fridge, lights). **Never "backfeed" a generator through a dryer plug** — it can kill line workers and burn the house.

Making propane last

- One gallon of propane holds about **91,500 BTU** of heat.
- Read the tank gauge (percent full). Gallons left ≈ tank size × gauge % ÷ 100. Example: 500-gal tank at 60% ≈ 300 gallons.
- **Write the reading daily** in the log (Chapter 13). After 3 days we'll know our burn rate: *gallons used per day → days left.*
- Conserve:
- Set the thermostat lower (around 60–62°F) in the house, and keep the **babies' room warmer**.

- Close vents and doors to unused rooms.
- Cover windows at night (blankets, bubble wrap, foil-faced foam board); open south windows on sunny days.
- Roll towels against door gaps.
- Shut off the propane water heater if not needed (heat water on the stove for washing instead).
- Keep the **tank lid, regulator, and path to the tank clear of snow** so it can be refilled and the regulator vent doesn't block.
- **Order propane early in the fall** — in a crisis, delivery may stop. Don't let it go below about 30% going into winter storms.

2. Reconnecting the wood stove to our chimney

We have a chimney and an old stove in the basement. This is a **great backup — but only if it is done right.** A bad connection can cause a **chimney fire, house fire, or CO poisoning**. Do it **now, while professionals are available**, so it's ready when we need it.

Step 1 — Inspect the stove

- Look for a **metal label** on the back (UL or other listing, brand, model, required clearances). If it has a label, the **label and owner's manual rule**.
- Check for **cracks** in the firebox (cast iron or steel), warped plates, missing firebricks, broken door glass, worn door gaskets (should seal tight — test by closing the door on a dollar bill; it should be hard to pull out).
- If cracked or badly worn: **replace it.** A new **EPA-certified stove** burns cleaner, uses less wood, and makes less creosote. *(EPA Burn Wise)*

Step 2 — Get the chimney inspected and swept

Hire a **CSIA-certified chimney sweep** (Chimney Safety Institute of America) for a **Level 2 inspection** (they often use a camera). They will check:

- **Cracks, missing mortar, or blockages** (bird nests are common).
- **Creosote** buildup — the black, tarry fuel that causes chimney fires.
- **The flue liner.** Many older masonry chimneys have **cracked clay tiles or no liner at all**. Stoves often need a **stainless-steel liner sized to the stove's flue outlet** (for example a 6-inch stove → a 6-inch liner). An oversized flue drafts poorly and builds creosote fast.
- **Is this flue shared** with the furnace or water heater? **A wood stove must have its own flue** unless the manufacturer and code specifically allow otherwise. Many propane appliances vent through the chimney — the sweep must check this.
- **Chimney height:** the top should be at least **3 ft above where it exits the roof** and **2 ft higher than anything within 10 ft** (the "3-2-10 rule"). Too short = poor draft and smoke.
- **Cap and spark arrestor** — important on the windy, dry plains.

Step 3 — The connection (stovepipe/connector)

- The pipe from the stove to the chimney is called the **connector**. It enters the chimney through a **thimble** (a clay or metal sleeve set in the masonry).
- **Single-wall stovepipe** needs **18 inches** clearance from combustibles (wood, drywall, joists) unless the stove or pipe label says otherwise. **Double-wall (close-clearance) pipe** allows less — follow its label. *(NFPA 211)*
- Keep the connector **short**, with few elbows. Horizontal runs should slope **up toward the chimney** (commonly about ¼ inch per foot).
- Crimped (male) ends point **down toward the stove** so creosote drips back inside.
- Fasten each joint with **3 sheet-metal screws**.
- **Never** run stovepipe through a wall, floor, or ceiling by itself. It must go through a listed thimble or wall pass-through.
- **Never connect into a fireplace opening without a proper liner and connection** — the old fireplace flue is usually far too big and unsafe.

Step 4 — Clearances and floor

- **Labeled stove:** use the clearances on the label/manual.
- **Unlabeled (unlisted) old stove:** NFPA 211 requires large clearances — **36 inches** from the stove to combustible walls/materials unless protected with approved shielding. Get the sweep or installer to confirm.
- **Floor:** a **noncombustible hearth pad** (stone, tile on cement board, or a listed pad) sized per the manual. A concrete basement floor usually works — but check that nothing combustible is under or near it.
- Nothing stored within 3 ft: no firewood stacks, laundry, kids' toys, or propane bottles.

Step 5 — Permits, insurance, test fire

- Ask **Morgan County building department** if a permit/inspection is required.
- **Tell our home insurance** — some require a professional install.
- **Draft test:** with a window cracked, hold a lit match or incense near the open stove door — smoke should be pulled **in**. If it's not, warm the flue first by burning a twist of newspaper. If it still smokes into the room, **stop and call the sweep.**
- Do a few small test fires **with CO alarms running** and someone watching.

Using the stove safely

- Burn only **dry, seasoned firewood** (split and dried at least 6 months, under 20% moisture — a cheap moisture meter checks this). *(EPA Burn Wise)* Wet wood = smoke, little heat, lots of creosote.

- **Never burn:** trash, plastics, treated or painted wood, pallets with "MB" stamp, cardboard boxes in bulk, or gasoline/charcoal lighter fluid to start it.
- Keep a **fireplace screen/safety gate** around the stove — **toddlers and the 6-year-old must never be near it.** A stove can cause severe burns in a second.
- **Ashes:** put in a **metal can with a lid**, set outside on dirt/concrete, away from the house. Ashes stay hot for days.
- **Sweep the chimney** at least once a year; check monthly when burning heavily.
- **Chimney fire signs:** loud roaring, popping, dense smoke, hot chimney. **Close the stove air controls, get everyone out, call 911.**
- Keep an **ABC fire extinguisher** near the stove and in the kitchen.

Firewood on the plains

Trees are scarce around Wiggins, so **stock up now**. Firewood is sold by the **cord** (a stack 4 ft × 4 ft × 8 ft = 128 cubic feet). How much we need depends on the house, stove, and winter — heating a whole house all winter can take **several cords**. Heating **one room** as an emergency backup takes much less. **Track it:** note how much one cold week uses, then multiply.

- Store wood **off the ground, covered on top, open on the sides**, at least 30 ft from the house when possible (fire safety), with a few days' supply closer.
- Scrap lumber (untreated), downed cottonwood along the river (with landowner permission), and tree trimmings can supplement.

If there were no chimney

(Not our case — we have one.) Never improvise a vent out a window or wall. A stove with no chimney needs a **professionally installed Class A insulated chimney pipe**.

3. Backup propane heaters

- **Only** use heaters **labeled for indoor use**, with an **oxygen depletion sensor (ODS)** and tip-over shutoff. Follow the manual about ventilation (most require a cracked window).
- Use with a **CO alarm**, never while sleeping, never near curtains or kids.
- **Never use for heat indoors:** gas oven or stovetop, charcoal or propane grills, camp stoves, patio heaters, kerosene "torpedo" heaters, or a generator. *(CDC)*
- Store 1-lb and 20-lb propane cylinders **outside**, never in the house or basement.

4. No heat at all — the "one warm room" plan

1. **Choose one room** — ideally an **interior room** with few windows, near the wood stove if it is running (the **basement** can work well: the earth around it holds steady temperature, and it's also our tornado shelter).
2. **Move everyone in**, including Grandma and Grandpa. Shared body heat matters.

3. **Insulate it:** hang blankets over doors and windows, put rugs/cardboard on the floor, tape plastic sheeting over windows.
4. **Make a "tent" inside the room:** a table covered with blankets, or a small camping tent, holds heat around sleepers.
5. **Layers:** wool or synthetic, not cotton. Hats, dry socks, mittens. Change damp clothes right away.
6. **Eat and drink:** the body makes heat from food. Warm drinks help (no alcohol — it makes hypothermia worse).
7. **Babies:** see Chapter 4 for safe warmth. Never put a baby to sleep with loose blankets, pillows, or in an adult bed.
8. **A tent inside the room** (The Prepared, after the 2021 Texas blackout): set up a camping tent with its rainfly on, pile blankets over it, and sleep inside. A person gives off roughly **330 BTU of heat per hour** — a small enclosed space traps it.
9. **Shovel paths often** during a storm (to the propane tank, coop, hutches, woodpile) rather than all at once at the end.
10. **Hot water bottles** (warm, not hot — wrap in a towel; never against a baby's skin).

5. Keeping an Insulated Garage Above Freezing With No Power (Protecting Stored Water)

The situation: the power is out, there's no service of any kind, and the garage heater is dead. The garage is still insulated, and **our stored water is in there.** If that water freezes solid, containers can split and we lose it. Here is how to protect it, **in order of what matters most.**

> **The big idea:** we don't need to heat the whole garage. We only need to keep **the water** from freezing solid. Pack it tightly, insulate it, and heat (if at all) only a small space around it.

Step 1 — Use the water's own "thermal mass"

Water holds a lot of heat and gives it up slowly. **A big container stays unfrozen much longer than a small one**, and a tight group of containers longer still, because each one shelters its neighbors. Water also **gives off a large amount of heat as it starts to freeze**, which slows the freezing of the rest. That is why an old trick for root cellars is to set a tub of water inside — the water protects the vegetables. *(Extension root-cellar guidance)*

As a rough guide: in a cold snap, a single gallon jug in an unheated space can freeze overnight, while a packed group of 55-gallon drums in an insulated garage can take **days** of hard cold to freeze solid. That gives us time.

- **Group all containers together** in one block — touching, no gaps. Stack jugs tightly; put the **biggest containers in the middle**.
- **Pick the warmest spot:** usually against the **wall shared with the house**, away from the overhead door and outside walls.

- **Get them off bare concrete** — especially near the door and the edges of the slab, which get very cold. Set them on **pallets with rigid foam board on top**, or at least on layers of cardboard or a wood floor.
- **Wrap the whole block:** blankets, sleeping bags, old comforters, foam board, or **stacked straw or hay bales** around the sides and on top. Insulation slows heat loss — and the water's stored heat keeps the inside of the block warmest.

Step 2 — Build a "room within the room"

If a cold spell will last more than a day or two, build a small enclosure around the water so there is only a little space to keep above freezing:

- **Hay or straw bales** stacked two high around the water make thick, cheap walls (we already buy bales for the rabbits). Cover the top with plywood and blankets, or a tarp over foam board.
- Or a frame of 2×4s or furniture with **foam board** panels, or a **tarp tent** draped over and weighted at the floor, with blankets on top.
- **Leave out any fuel-burning heater from a hay, straw, or tarp enclosure.** Those burn fast. Inside the enclosure, use only **heat that can't start a fire**: hot water jugs (Step 5), the water's own mass, and animal or compost heat.

Step 3 — Containers: leave room to freeze

- Water **expands about 9% when it freezes.** Leave roughly **a tenth of each container empty** (headspace) so it can expand without splitting.
- **Sturdy rigid containers** (food-grade drums, thick 5–7 gallon jugs, water bricks) survive better than **thin milk-type jugs**, which crack easily.
- **If water freezes, it's still usable.** Let it thaw slowly (bring a jug into the house), check the container for cracks, and use it. Treat it as you would any stored water (Chapter 2) if you're unsure.
- Keep a few **small containers in the house** as a backup — they're easier to thaw and carry.

Step 4 — Stop the leaks and use the sun

- **Seal drafts:** weatherstrip or stuff rags into gaps around the **overhead door** (bottom seal, sides, and between panels), the side door, and any vents that aren't needed. Foam board panels can be cut to fit inside the overhead door panels.
- **Cover windows at night** with foam board or blankets; **open them on sunny days.**
- **Passive solar heat:** on clear winter days, sun through a **south or west window** can warm a garage noticeably. Put **dark-painted water containers or a black-painted barrel** where the sun hits them — they soak up heat in the day and give it back at night. **Uncover in the morning, cover at dusk.** Our plains winters have many sunny days — use them.

Step 5 — Carry heat out from the house

If the **house wood stove** is running (Section 2), it can heat the garage water too — safely:

- Heat water in big pots on the stove, pour it into **sturdy jugs or a bucket with a lid**, and set them inside the insulated water block or enclosure. Refill each morning and evening.
- **Rough idea of the power:** a few gallons of very hot water cooling down to near freezing gives off roughly as much heat as a small propane heater on low runs in an hour. It's not a lot, but **several trips a day can keep a well-insulated water block from freezing.**
- Use **hot (not boiling) water** in containers rated for it, carry with gloves, and keep it away from kids.

Step 6 — Fuel-burning heat: ONLY the safe way

> ⚠ **Never use these inside the garage — not even with the door cracked:** a **running vehicle**, a **generator**, a **charcoal or propane BBQ grill**, a **camp stove**, or a **kerosene or "torpedo" construction heater**. They make **carbon monoxide** — a gas you can't see or smell that can kill in minutes and **can seep into the house** through the shared wall and door. *(CDC, NFPA)*

An indoor-rated portable propane heater (the "Mr. Heater Buddy" type) is the one fuel-burning option made for enclosed spaces:

- It must be **labeled for indoor use** with a **low-oxygen shutoff (ODS)** and a **tip-over shutoff**.
- **Ventilate as the manual requires** — usually a window or door cracked open a set amount. Read the manual **now** and write the requirement here: _____.
- **A battery CO alarm in the garage is mandatory** — and one on the house side of the shared door.
- **Remove gasoline, cans of paint, thinner, and solvents** from the garage or keep them far from the heater — their vapors can ignite. **Keep 3 feet clear** around the heater (no hay, cardboard, tarps, or blankets). *(NFPA)*
- **Never leave it running while no one is checking it**, and **never while sleeping** in the garage.
- **Propane:** use **1-lb cylinders** inside as the manufacturer allows. Larger 20-lb tanks and **our big tank out back must stay outside** — only use the heater maker's approved hose and fuel filter, following its directions. **Never run a homemade line off the main tank** — that's a job for the propane company. Store spare cylinders **outside**.
- **Rough guide:** a 1-lb cylinder lasts only **hours**, not days, so plan to use the heater in **short runs on the coldest nights**, aimed at the insulated water block — not to heat the whole garage.

A wood stove in the garage? Only with a **properly installed, listed chimney, the required clearances, and a non-combustible floor pad** — the same rules as Section 2 *(NFPA 211)*, and no gasoline or solvents stored in the garage. Not something to improvise in a crisis.

Step 7 — Small heat sources (honest limits)

- **Candles:** a handful of candles gives off only about as much heat as a person or two — **minor**, and a fire risk. **"Terra-cotta flower-pot heaters" don't create extra heat**; they only spread out the candle's heat. Not worth the fire risk near hay, tarps, or blankets.
- **Animals: rabbits and chickens give off body heat**, and a coop or hutch built against the water block adds a little warmth. But animals also bring **moisture, ammonia, and dust** (and mice — Chapter 9, hantavirus). Never put them near a fuel-burning heater.
- **Compost heat:** a big, active pile of **manure and bedding** (rabbit manure + straw is ideal) can heat to well over 100°F inside. *(CSU Extension)* It needs a **large volume — about a cubic yard or more** — and moisture. A compost bin built **against the insulated water block** can add slow, steady heat; a small pile does little. Watch for ammonia.
- **People:** working in the garage adds a little heat — but don't stay in a closed garage with any fuel-burning heater.

Step 8 — Watch the temperature

- Put a **min/max thermometer** (shows the coldest it got overnight) or a **battery wireless thermometer** with the sensor **inside the water block** and read it from the house.
- **Rough guide:** if the air inside the enclosure stays **above freezing**, the water is safe. If it drops below freezing for a night, the big containers are usually still fine — act before it happens two or three nights in a row.
- When a hard cold front is forecast on the NOAA radio, **top up the insulation and plan hot-water trips ahead of time.**

Garage checklist — stock this now

- [ ] Pallets + rigid foam board (2" is a good start) for under and around the water
- [ ] Old blankets, comforters, sleeping bags, or moving blankets for wrapping
- [ ] Straw/hay bales (doubles as rabbit bedding), tarps, plywood for a lid
- [ ] Sturdy rigid water containers; fill to leave about a tenth empty
- [ ] Overhead-door bottom seal, weatherstrip, rope caulk, foam board for windows
- [ ] Black paint or dark containers for a sunny window spot
- [ ] Insulated jugs or buckets with lids for carrying hot water; heavy gloves
- [ ] Indoor-rated propane heater with ODS + 1-lb cylinders (stored outside) + manual
- [ ] **Battery CO alarms**: one in the garage, one on the house side of the shared door
- [ ] ABC fire extinguisher in the garage
- [ ] Min/max or wireless thermometer + spare batteries
- [ ] A place outside to move gasoline, paint, and solvents away from any heater

Summer heat

Summers here can be very hot and dry. Babies and older adults overheat fast.

- Keep the house closed and shaded in the day, open it at night.
- Use the basement as the cool room.
- Wet cloths, cool (not cold) baths for kids, plenty of fluids.
- **Never leave a child or pet in a car.**
- Heat stroke (hot, confused, not sweating) = **911**.

Do it now

- [ ] Battery **CO alarms** on every level and sleeping area, both houses. Smoke alarms tested.
- [ ] **CSIA-certified sweep: Level 2 chimney inspection** + liner recommendation.
- [ ] Inspect/repair or replace the stove; professional connection; ask county about permit; tell insurance.
- [ ] Hearth pad, stove gate, metal ash can, ABC extinguisher, moisture meter.
- [ ] Buy firewood (dry) and store it right.
- [ ] Electrician: generator inlet + transfer switch/interlock (furnace, well, fridge).
- [ ] Fill propane in early fall; record gauge.
- [ ] Wool blankets, sleeping bags rated for cold, plastic sheeting, duct tape.

CHAPTER 4

Babies & Kids

Our kids are the most vulnerable people in the family. Babies **get cold, dehydrated, and sick faster** than adults, and they can't tell us. This chapter is deliberately careful. **If services are running, call the pediatrician or 911 when in doubt. If they aren't, Chapter 9 covers caring for a sick baby at home.**

Our kids at a glance

Kid	Feeds on	Biggest risks in an emergency
Newborn (born late Sep 2026)	**Breast milk only**	Cold, dehydration, fever/infection, unsafe sleep
1-year-old	Formula → moving to whole milk	Running out of formula/milk, choking, dehydration
2-year-old	Regular milk + table food	Dehydration from diarrhea, choking, burns (stove)
6-year-old	Regular food	Fear/stress, wandering, burns, cold exposure outdoors

The newborn — breastfeeding

Breastfeeding is the safest emergency food in the world: always the right temperature, no water needed, no bottles to clean, and it protects against infection. **Protect Mom's ability to nurse.**

- Newborns typically nurse **8–12 times per 24 hours**. *(AAP)* Keep nursing on demand, even during stress — the more baby nurses, the more milk is made.
- **Stress does not usually dry up milk**, but it can slow let-down. Skin-to-skin, a warm quiet spot, and a calm helper help.
- **The newborn needs NO water, juice, tea, or food** — only breast milk. Water can be dangerous for young babies.
- **Is baby getting enough?** After the first week: about **6 or more wet diapers a day**, regular stools, swallowing sounds while feeding, steady weight gain, alert when awake.
- Get a **manual breast pump** (no power needed) and learn **hand expression** — in case Mom and baby are briefly apart, or Mom is engorged.
- Pumped milk storage *(CDC)*: room temperature up to **4 hours**; refrigerator up to **4 days**; freezer about **6 months best (up to 12)**.

Feeding Mom (very important)

- A breastfeeding mother needs about **450–500 extra calories a day**. *(CDC)* **She eats first** when food is short — her milk feeds the baby.
- **Drink to thirst** — a simple habit is a full glass of water every time she nurses. Dark urine = drink more.
- Keep taking a **prenatal vitamin**; stockpile several months.
- Eggs and rabbit are great protein for her — add fat and carbs (see Chapter 6 on "rabbit starvation").
- If Mom gets sick (cold, flu, stomach bug), she can usually **keep breastfeeding** — her antibodies pass to the baby. Check any medicine with the doctor/pharmacist or LactMed (a National Library of Medicine database).

Backup if Mom can't nurse

If Mom were injured, very ill, or separated, the newborn would need **infant formula**. Keep a small backup supply of **ready-to-feed infant formula** (no water needed) and disposable nipples — but don't use it routinely, because skipping feeds lowers Mom's milk supply. Ask the pediatrician which formula.

The 1-year-old — formula and moving to milk

- The American Academy of Pediatrics says most children can switch from formula to **whole cow's milk at about 12 months**, about **16–24 oz (2–3 cups) a day**, plus solid foods. **Toddler/"transition" formulas are not needed.** *(AAP / HealthyChildren.org)* Talk to the pediatrician about switching **now**, before an emergency — it makes us less dependent on formula.
- Too much milk (over about 24 oz a day) can fill them up and lower iron. Offer iron-rich foods: **eggs, meat (including rabbit), beans, iron-fortified cereal.**

If still on formula — stockpile math

1. Check the can: it says **"makes about ___ fl oz."**
2. Daily ounces × days ÷ ounces per can = cans needed. *Example: 24 oz/day × 30 days = 720 oz. If a can makes 90 oz → 8 cans for a month.*
3. Rotate: use the oldest first, watch expiration dates.

Formula safety — never break these rules

- **NEVER make homemade formula.** It can cause serious illness and death. *(FDA, AAP)*
- **NEVER water down formula** to stretch it. It can cause dangerous low sodium (seizures) and malnutrition. *(AAP)*

- **NEVER give an infant under 12 months cow's milk, goat milk, or plant milk** as a formula substitute.
- Mix **exactly** as the label says.
- Use **safe water** (bottled, or boiled then cooled). **Ready-to-feed** is best when water is uncertain.
- Throw out prepared formula after the time on the label (often 1 hour once baby starts drinking).

The 2-year-old — milk and food

- Keeps drinking milk + eating family food. Ask the doctor whether to stay on whole milk or switch to lower-fat after age 2.
- **Shelf-stable milk:** stock **UHT (boxed) milk** and **powdered whole milk** for when fresh milk runs out. Once opened, refrigerate/use quickly.

Safe foods from our farm for the little ones

- **Eggs:** great food for all the kids except the newborn. **Cook fully** — firm yolks and whites — for young children. *(Eggs can be introduced to babies around 6 months; the newborn is far from that.)*
- **Rabbit meat:** cook thoroughly; shred or mince finely; **remove every small bone** — rabbit bones are thin and sharp.
- **Choking hazards for under 4:** whole grapes, nuts, popcorn, hot dog rounds, hard raw vegetables, chunks of meat, spoonfuls of peanut butter. Cut food into small, soft pieces.
- **No honey** for babies under 1 year (botulism).
- **Wash hands after handling chickens or eggs** — chickens commonly carry Salmonella. Kids under 5 should not kiss or snuggle chickens. *(CDC)*

Keeping babies warm (and safe)

- Dress the baby in **about one more layer than an adult** would wear to be comfortable. *(AAP)*
- **Feel the chest or back of the neck** — warm and dry is right. Cold hands/feet alone are normal; sweaty or flushed = too hot.
- **Skin-to-skin** with an awake adult, covered by a blanket, is an excellent way to warm a baby.
- **Safe sleep, every sleep** *(AAP)*: on the **back**, on a **firm, flat surface** (crib, bassinet, or play yard) **in the same room** as parents, with **no loose blankets, pillows, or bumpers**. Use a **wearable blanket/sleep sack** and footed pajamas for warmth instead of loose blankets. Don't let baby sleep in a car seat outside the car, a swing, or an adult bed or couch.
- Keep babies **away from the wood stove and space heaters** — and keep the sleep space away from them too.
- **A baby who is cold to the touch, unusually quiet, floppy, or not feeding = emergency.**

Diapers and hygiene

- Rough daily use: **newborns about 8–12**, older babies/toddlers fewer. Stock **at least a month** of each size the kids will be in (sizes change fast!).
- **Backup: cloth diapers** (prefolds + covers) and a bucket with lid. Washing needs water and heat — plan for it.
- Wipes, diaper cream, garbage bags, and hand sanitizer.
- Bury or bag soiled diapers away from water and gardens; never in chicken compost.

Dehydration — know the signs

Kids can get dangerously dry from **diarrhea, vomiting, fever, or heat**.

Warning signs:

- Fewer wet diapers (or none in 6+ hours for a baby), dark urine
- Dry mouth, **no tears** when crying
- **Sunken soft spot** on top of a baby's head, sunken eyes
- Very sleepy, floppy, or very fussy

What to do: keep breastfeeding/feeding; give **oral rehydration solution** (store-bought such as Pedialyte) in small sips for kids over infancy as directed on the package and by the doctor. **A newborn with any of these signs needs a doctor right away if one can be reached; if not, see Chapter 9 — feed often, give fluids by spoon, and track diapers.**

Fever

- **Any baby under 3 months with a temperature of 100.4°F (38°C) or higher is an emergency — if services are running, go to the ER immediately.** *(AAP)* That is our newborn right now. **With no help coming, follow "A fever in our newborn" in Chapter 9.**
- Get a **digital thermometer** (rectal is most accurate for babies).
- Ask the pediatrician **now** for the correct dose of acetaminophen (and ibuprofen for 6+ months) by weight, and write it in the Home Worksheet. **Never give aspirin to children.**

The 6-year-old (and keeping kids steady)

- Kids handle hard times better with **routine, honest simple explanations, and jobs**. Give the 6-year-old real jobs: gathering eggs, feeding rabbits, carrying kindling (supervised), being "radio helper."
- Teach: our address, parents' names and phone numbers, the meeting places, "stay away from the stove and the road," and to **never go outside in a blizzard alone**.
- Keep books, crayons, cards, and favorite toys in the emergency kit.

- Watch for stress: bedwetting, nightmares, clinginess. It's normal — extra hugs and routine help.

Do it now

- [ ] Ask the pediatrician: switching the 1-year-old to whole milk; medicine doses by weight; what to keep in a baby kit.
- [ ] Manual breast pump + bottles + milk storage bags.
- [ ] Small backup of **ready-to-feed infant formula** (newborn) + 1 month of the 1-year-old's formula/milk plan.
- [ ] UHT boxed milk and powdered whole milk.
- [ ] 1 month of diapers in the next sizes + cloth backup, wipes, cream.
- [ ] Prenatal vitamins (3+ months) for Mom.
- [ ] Digital thermometer, infant/children's acetaminophen, oral rehydration solution.
- [ ] Wearable blankets/sleep sacks, footed pajamas, a portable play yard for the warm room.
- [ ] Stove gate.

CHAPTER 5

Food: Eggs & Chickens

Our hens give **6–12 eggs a day** — that's real food: each large egg has about **70 calories and 6 grams of high-quality protein** *(USDA)*. Nine eggs a day ≈ **630 calories and 55 g protein** for the family. The challenge is **keeping the hens fed and laying when we can't buy feed, especially in winter.**

How much hens eat and drink

- A laying hen eats roughly **¼ lb of feed a day** (more in cold weather). *(University Extension, typical figure)*
- 15 hens ≈ 3¾ lb/day ≈ **110 lb per month**.
- A hen drinks roughly **a pint of water a day**, more in heat. **No water = no eggs within a day or two.**
- Laying hens need feed of about **16% protein** plus **calcium** for strong shells. *(Extension)*

Stockpile now: several months of **layer feed** in metal cans with lids (mice!), plus **oyster shell**. Feed goes stale — rotate, use oldest first, keep dry. **Never feed moldy feed.**

Why hens stop laying in winter

1. **Short days.** Hens lay best with about **14–16 hours of light**. In December, Wiggins gets only about 9½ hours of daylight. *(Extension)*
2. **Molting** in fall — they stop to regrow feathers (needs extra protein).
3. **Cold stress, not enough feed or water.**
4. **Age** — production drops after the second year.

What helps:

- **Add light** to make about 14 hours total: a low-watt LED on a timer, turned on **early morning** (so they go to roost at natural dusk). If power is out, a small solar light kit can do this.
- **Raise protein** during molt (insects, cooked egg, meat scraps — see below).
- **Keep water unfrozen.** Heated waterer base if power; otherwise swap waterers 2–3 times a day with warm water.
- **Plan to replace hens:** hatch or raise new pullets each spring so we always have young layers. A rooster lets us hatch our own (a broody hen or incubator).

Winter coop care on the plains

- **Ventilation without drafts:** vents high up, no wind blowing on roosting birds. Damp air causes **frostbite on combs** more than cold does.

- **Deep litter:** keep adding dry straw/pine shavings; it composts slowly and adds a little warmth.
- **Avoid heat lamps** — they are a leading cause of coop and barn fires. Chickens handle cold well if dry and out of the wind.
- **Wind block:** straw bales, tarps, or plywood on the north and west sides.
- Close them in **securely at night**: coyotes, foxes, raccoons, skunks, and owls are all around Wiggins. Hardware cloth, not chicken wire.

The "Raken" house — rabbits over chickens (Joel Salatin)

At Polyface Farm, Joel Salatin's family hangs **rabbit cages above a floor where hens run** — the "Raken" (rabbit + chicken) house. Rabbit droppings and urine fall into **deep carbon bedding** (wood shavings, sawdust, leaves, chopped straw). The hens scratch it constantly, which keeps it from smelling, turns it into compost, and **breeds worms and bugs that the hens eat — free protein all winter.**

Salatin's deep-bedding rules:

- Start with **at least 8 inches, preferably 12+**, of carbon bedding.
- **Add more carbon whenever it smells.** Long hay/straw mats — chop it or add a little at a time.
- A dirt floor drains best. Good ventilation, especially in warm weather.
- Toss scratch grain on the bedding to get the hens working it.
- The finished compost is excellent garden fertilizer in spring.

This fits our place perfectly: it **puts Daniel's two animals together** so the rabbits help feed the chickens without killing a single rabbit, and it keeps both in one sheltered, easier-to-water building in winter. *(Keep rabbit cages high and solid enough that hens can't roost on them and soil the rabbits.)*

Feeding chickens when feed runs out

Chickens are **omnivores** — they eat grains, greens, seeds, bugs, and meat. Mix several of these:

Grains (energy)

- **Whole or cracked wheat, corn, milo (sorghum), barley, oats.** We live in grain country — buy or trade directly from local farmers and elevators **now**, by the bag or ton. Grain stores for years if dry and sealed.
- Corn alone is **low in protein** — hens will lay poorly on corn only. Add protein.

Protein (eggs need protein)

- **Insects** (see the bug farms below).
- **Cooked eggs** (scrambled, with shell crushed in) — especially cracked or extra eggs.

- **Cooked rabbit scraps and organs** (see below).
- Fish scraps, cooked; milk/whey or cottage cheese in small amounts.
- **Dried peas** or other legumes — **cooked**. Raw dry beans are toxic.

Calcium (shells)

- **Oyster shell** free-choice; or **our own eggshells** — rinse, **bake dry**, crush fine, offer in a separate dish.

Greens (vitamins, especially in winter)

- Stored **pumpkins, squash, cabbage, beets, carrots** (grow extras in summer for the chickens).
- **Alfalfa hay leaves** (shake the leafy bits out of a bale).
- **Sprouted grain "fodder"**: soak wheat or barley overnight, spread in trays with drain holes, rinse twice a day, feed in **about 6–8 days** when it's a green mat. *Honest note:* sprouting does **not** add calories (it uses some up), but it gives **fresh greens and vitamins in winter** and hens love it. Needs room temperature and water.

Kitchen scraps

- **OK:** vegetable and fruit scraps, cooked rice/pasta/oats, bread (small amounts), cooked meat scraps, garden waste.
- **Not OK: moldy or rotten food**, **avocado** (pits/skins), **raw or dried beans**, **chocolate**, very **salty** food, **green potato peels**, onions in large amounts, coffee grounds.

Bug farms — protein we can raise indoors in winter

Mealworms (easiest for winter, indoors)

- Plastic bins (with ventilation holes) of **wheat bran or rolled oats** a few inches deep.
- Add **slices of carrot or potato** for moisture (replace before mold).
- Keep at **room temperature** — the life cycle (beetle → egg → worm → pupa → beetle) takes a few months, slower when cool.
- Start in **late summer** so the colony is big by winter. Harvest larger worms; leave enough to breed.
- Start with a few thousand bought live mealworms.

Black soldier fly larvae (BSFL) — best in summer

- Larvae eat almost **any food waste and manure** and are high in protein and fat. The mature larvae crawl out on their own into a bucket ("self-harvesting" bins).

- **They need warmth** (they slow down a lot in cool weather) — great from late spring to early fall outdoors. Over winter, they're hard to keep without a heated space. Plan for summer production and **dry the extra larvae** (in an oven or dehydrator) to store for winter.

Earthworms (red wigglers)

- A worm bin with shredded cardboard, kitchen scraps, and rabbit manure (rabbit manure is great for worms). Keep it above freezing — basement works.

Daniel's idea: rabbits as chicken food

Yes — this works, with the right approach. Chickens love meat and it's real protein, especially for winter laying and molting.

Do it this way:

- **We eat the rabbit meat.** Feed the chickens the parts we don't eat: **organs we're not using, heads, scraps, fat trimmings, and bones (cooked, then cracked)**.
- **Cook it first.** Boil scraps and organs until fully cooked. Cooking kills parasites and germs (including **tularemia**, which rabbits can carry) and keeps the hens from developing a taste for raw meat that can lead to egg- or flock-pecking.
- **Never feed spoiled or sick-animal meat.** Bury or burn any rabbit that died of illness.
- Feed scraps the same day, a modest amount, alongside grain — not as the only food.

The energy trade-off (important): every time food passes through an animal, **most of its energy is lost**. A rabbit turns hay into meat inefficiently; a hen turns rabbit meat into eggs inefficiently. **Feeding whole rabbits to chickens to get eggs yields far fewer calories than eating the rabbit ourselves.** So the best plan is: **people eat the meat; chickens get the leftovers.** That's a win twice.

About the reverse: **rabbits are strict plant-eaters** — never feed chicken or any meat to rabbits.

Keeping eggs

- **Unwashed** eggs straight from clean nests keep their natural coating. The USDA recommends refrigeration for safety; in an outage keep eggs in the coolest place possible and use oldest first.
- **Freeze extra eggs:** crack, beat lightly, freeze in portions (ice cube trays). Don't freeze in the shell.
- Traditional preservation (like "water-glassing" in pickling lime) is popular but **not tested/approved by the USDA** — use at your own risk; discard any egg that smells off or floats suspiciously.
- **Cook eggs fully** for the kids, Grandma and Grandpa, and Mom.

Flock health and bird flu

- **Avian influenza (bird flu)** has hit Colorado poultry and dairies, including in nearby Weld County. Keep wild birds out of feed and water, don't touch sick or dead wild birds, wash hands, use dedicated boots for the coop.
- **Sudden deaths, swollen heads, purple combs, or a big drop in eating/laying:** don't eat those birds or eggs; report to the **Colorado Department of Agriculture State Veterinarian's office**.
- If you butcher an old hen for soup: only healthy birds, cook thoroughly.

Do it now

- [ ] Stockpile 3+ months of layer feed (sealed metal cans) + oyster shell.
- [ ] Line up a local farmer/elevator for bulk wheat, corn, or milo.
- [ ] Coop light on a timer; heated waterer (+ manual backup).
- [ ] Start a mealworm bin now; plan BSFL bin for next summer.
- [ ] Grow extra squash/pumpkins/beets next summer for winter greens.
- [ ] Get a rooster or plan spring chicks to keep young layers coming.
- [ ] Predator-proof the coop with hardware cloth.

CHAPTER 6

Rabbits & Feeding Eight People

Rabbits are one of the best survival meat animals: quiet, small, fast-breeding, and they turn hay and greens into meat. **But you cannot live on rabbit alone** — the meat is very lean. This chapter covers raising, butchering, storing, and *how rabbit fits into a full diet for 8 people*.

Rabbit basics

Fact	Typical figure *(Extension guides, meat breeds)*
Pregnancy (gestation)	about **31 days** (28–35)
Litter size	often **6–8 kits** (varies a lot)
Weaning	about **4–8 weeks**
Butcher age ("fryers")	about **8–12 weeks**, around 4–5 lb live weight
Dressed meat	a bit over **half** the live weight (with bone)
First breeding (medium breeds)	does around **5–6 months** old

A breeding plan

- Keep a few good **does** (females) and **1–2 bucks** (males), housed separately.
- **Always take the doe to the buck's cage** (does defend their space).
- Put a **nest box** with straw/hay in at **day 27–28**. The doe pulls fur to line it.
- A moderate, sustainable plan: rebreed the doe when kits are about 4–6 weeks old, giving about **4–5 litters a year** per doe. Rest does that look thin.
- Keep records: doe, buck, breeding date, kindling date, number of kits, weaned, butchered.
- **Avoid inbreeding:** bring in an unrelated buck every year or two (trade with neighbors).

Rough output: 4 does × 4–5 litters × ~6 kits raised ≈ **about 100 fryers a year** — roughly **one rabbit every 3–4 days**. Great protein, but not enough calories alone for 8 people (see math below).

Feeding rabbits — especially in winter

- **Unlimited grass hay** (or grass/alfalfa mix) is the base of the diet. **Stock hay before winter** — it's cheap here in fall and scarce after a bad winter. Track how fast we use it.
- **Pellets** if available (stockpile in sealed cans). If pellets run out: hay + **oats/barley** (small amounts) + greens.

- **Safe greens and extras:** dandelions, plantain, clover, grass, carrot tops, kale, cabbage (small amounts), sunflower heads, sprouted grain fodder, tree branches (apple, willow — rabbits love to chew).
- **Avoid:** moldy hay, lawn clippings that have fermented, potato leaves, rhubarb, nightshade, avocado, onions, ice-cold or spoiled food, and **any meat**.
- **Water is the #1 winter problem.** Bottles freeze and crack — use **heavy crocks** and swap with warm water 2–3 times a day. Nursing does drink a lot.

Winter and summer care

- Rabbits handle **cold** well if **dry and out of the wind**. Wrap hutches on the north/west, add a roof overhang, and give extra hay to burrow into.
- **Winter kits are at risk** of freezing. Give a deep, well-bedded nest box; move the nest box (or the doe's cage) to a sheltered spot like a garage or barn in a cold snap. Or plan breeding so most kits are born in milder months.
- **Heat is the bigger killer** — rabbits suffer above about 85°F. In summer: shade, airflow, frozen water bottles to lean against.

Housing tip

Consider the **Raken house** (Chapter 5): rabbit cages hung above the chicken floor. The rabbits' manure feeds the deep bedding the hens scratch for bugs, and one building is easier to keep out of the wind and keep water thawed.

Butchering (humane and clean)

1. **Withhold feed** about 12 hours before (keep water).
2. Dispatch quickly and humanely (learn from an experienced person or a CSU Extension class — **practice now**).
3. Hang, skin, and gut on a clean surface. Keep the bladder and intestines intact.
4. **Check the liver:** it should be smooth and dark red. **White spots** may mean parasites/disease — throw out the liver; if the whole carcass looks unhealthy, discard it.
5. **Wear gloves.** Rabbits can carry **tularemia**, a bacterial disease. Domestic rabbits are low risk; **wild rabbits** acting sick or sluggish — don't handle or eat. *(CDC)*
6. Chill the meat quickly (cold water bath or fridge).
7. **Organs and scraps → cook for the chickens** (Chapter 5). Hides can be tanned; manure goes to the garden (rabbit manure can go straight on beds).

Storing rabbit meat

- **Fridge:** cook within 1–2 days.

- **Freezer:** months. Wrap tightly to prevent freezer burn.
- **Winter:** a cold snap is a natural freezer — but meat outside must be protected from animals and kept **solidly frozen**, not thawing in sun.
- **Canning:** meat **must be pressure-canned** using a tested USDA/Extension recipe. **Never water-bath can meat** — botulism risk.
- **Jerky:** USDA recommends heating meat to **160°F** before or after drying for safety.
- **Cooking:** cook rabbit thoroughly — to **160°F** internal temperature. *(USDA)*

⚠ "Rabbit starvation" (protein poisoning)

Rabbit meat is **very lean** — mostly protein, little fat. People who eat **only lean meat** for days to weeks get sick: **nausea, diarrhea, weakness, and constant hunger**, and can actually starve while eating plenty. Your body can only safely handle a limited share of calories from protein; the rest must come from **fat and carbohydrates**.

Always pair rabbit with:

- **Fat:** cooking oil, lard, butter/ghee, bacon grease, peanut butter, nuts, **egg yolks**.
- **Carbs:** rice, wheat/flour, oats, potatoes, beans, corn, squash.
- Example meal: rabbit stew with potatoes, beans, and a big spoon of lard or oil; eggs on the side.

Calorie math for our 8 people

These are **rough daily estimates** based on the USDA Dietary Guidelines' calorie tables (moderately active), adjusted for breastfeeding:

Person	Approx. calories/day
Daniel (physical work)	2,800
Wife (breastfeeding: +450–500)	2,500
Grandpa	2,200
Grandma	1,800
6-year-old	1,400
2-year-old	1,000
1-year-old (part from milk)	900
Newborn	breast milk only
Family total	**≈ 12,600 per day**

Work in the cold burns more. These are estimates, not prescriptions.

What the farm provides

- **Eggs:** ~9 eggs/day × 70 cal ≈ **630 cal/day** (less in winter).
- **Rabbit:** one fryer every 3–4 days. A fryer gives roughly 1,000 calories of meat. *(USDA lists raw rabbit at about 136 cal per 100 g.)* ≈ **300 cal/day**.
- **Together: about 1,000 calories/day — under 10% of what we need.** Eggs and rabbits are **excellent protein**, but **stored staples must supply most of the calories.**

What we'd need from storage (per month)

12,600 cal × 30 days ≈ **378,000 calories**. Roughly:

Staple	Approx. calories per lb (dry)	One month for us
Rice / wheat / flour / oats	~1,600	**~150 lb**
Dry beans / lentils	~1,550	**~40 lb**
Cooking oil / lard	~4,000	**~2 gallons** (≈ 15 lb)
Eggs + rabbit (farm)	—	~30,000 cal

Plus: salt, sugar/honey, powdered milk, canned vegetables and fruit, vitamins, spices, and comfort foods for the kids. See Chapter 7.

Do it now

- [ ] Count does, bucks, ages; start breeding records.
- [ ] Stock hay for the full winter + sealed pellets.
- [ ] Heavy water crocks for winter.
- [ ] Learn butchering hands-on (CSU Extension / experienced neighbor).
- [ ] Pressure canner (and a tested meat canning guide) or freezer space.
- [ ] Store fats: oil, lard, peanut butter.

CHAPTER 7

Food Storage, Gardening & Foraging on the Plains

Chapter 6 showed that **eggs and rabbits supply protein, but stored staples must supply most of our calories.** This chapter covers what to store, how to cook without power, how to garden in our short, windy, hail-prone season, and what we can realistically find, hunt, or fish around Wiggins.

Storage principles

1. **Store what we already eat**, and eat what we store (rotate: new in back, old in front).
2. **Cool, dark, dry, and rodent-proof.** Basement shelves are good. Off the floor.
3. Bulk grains and beans: **food-grade 5-gallon buckets** with **Mylar bags and oxygen absorbers**, or sealed metal cans. Stored this way, dry white rice, wheat, and dry beans keep for many years (Extension guidance).
4. Write the **date** on everything.
5. **Manual can opener** (two!).

How much — the LDS (Latter-day Saints) method

The Church of Jesus Christ of Latter-day Saints has taught home food storage for generations, and its method is simple and well tested:

1. **Start with one week** of the food you normally eat (what you buy for a day × 7).
2. Grow it to a **three-month supply** of normal foods, rotated.
3. Then build a **longer-term supply of basics that keep you alive: wheat, white rice, and beans**. Properly packaged and kept cool (75°F or below) and dry, these can last **30 years or more**. Cooking oil needs rotating every 1–2 years.

Their one-year "basics" per adult (Church guidance): **400 lb grains, 60 lb legumes, 16 lb powdered milk, 10 qt cooking oil, 60 lb sugar or honey, 8 lb salt**, plus 2 weeks of water. Children get a share: **3 and under 50%, 4–6 70%, 7–10 90%**, and they suggest using **next year's age** for growing kids. **Nursing babies share Mom's portion.**

For our family (4 adults + 6-year-old at 90% + two little ones at 50% each ≈ **5.9 adult shares**):

Basic	Per month	One year
Grains (wheat, rice, oats, corn, pasta)	~195 lb	~2,360 lb
Legumes (beans, lentils, split peas)	~30 lb	~355 lb
Powdered milk	~8 lb	~95 lb
Cooking oil	~5 qt	~59 qt (~15 gal)
Sugar or honey	~30 lb	~355 lb
Salt	~4 lb	~47 lb

These line up with our calorie math in Chapter 6. A year is a big goal — three months is a great start.

Warning (LDS/BYU guidance): only store **dry foods (about 10% moisture or less)** in airtight containers with oxygen absorbers — moist foods sealed without oxygen can cause **botulism**. Nuts, whole-wheat flour, brown rice, and jerky don't keep long-term.

A balanced pantry (starter targets for one month — multiply for more)

Group	Examples	Month target (approx.)
Grains	wheat berries + hand grain mill, white rice, oats, flour, pasta, cornmeal	150 lb
Beans/legumes	pinto, black beans, lentils, split peas	40 lb
Fats	vegetable oil, lard, shortening, peanut butter, nuts	2 gal oil + extras
Milk	UHT boxed milk, powdered whole milk	see Chapter 4
Canned	vegetables, fruit, tomatoes, soups, tuna, chicken	a can or two per person per day
Sugar/sweet	sugar, honey (not for newborn), jam	10–20 lb
Salt & seasoning	salt (also for preserving), spices, bouillon	10+ lb salt
Kids	baby cereal (iron-fortified), pouches, crackers, favorite snacks	as needed
Vitamins	multivitamins, prenatal vitamin	3+ months

Cooking without power

- **Wood stove top** (once reconnected) — great for pots of beans, stew, and heating water.
- **Propane camp stove or grill — OUTDOORS ONLY.** Never indoors or in a garage: carbon monoxide.
- **Thermal cooking:** bring beans/rice to a boil, then wrap the pot in blankets or put it in a cooler — it keeps cooking for hours with no fuel.
- Soak beans overnight to save fuel.

Fridge and freezer in an outage *(USDA)*

- Keep doors **closed**. Full freezer: ~**48 hours**; half full: ~**24 hours**. Fridge: ~**4 hours**.
- In winter, move food to coolers outside in the shade (protected from animals) when it is below freezing.
- **"When in doubt, throw it out"** — especially meat, dairy, eggs, leftovers above 40°F for more than 2 hours.

Gardening on the eastern plains

Our challenges: **short season, late spring and early fall frosts, hail, wind, alkaline soil, and limited water.**

- **Frost dates:** around Wiggins the average last spring frost is roughly **mid-May** and the first fall frost roughly **late September to early October**. Check **CSU Extension / Colorado Climate Center** for exact local dates.
- **Best calorie crops for us:** potatoes, winter squash and pumpkins (store all winter — and feed chickens), dry beans, corn, sunflowers (seeds for people, chickens, and rabbits).
- **Fast greens:** lettuce, spinach, kale, radishes, peas (cool season — plant early spring and again late summer).
- **Protect from hail and wind:** hail cloth or row covers on hoops, windbreak fences, raised beds.
- **Season extenders:** cold frames, low tunnels, a small hoop house — can stretch greens well into fall.
- **Water:** drip irrigation and thick mulch (straw). **Colorado law** lets homeowners collect rooftop rainwater in **up to two rain barrels totaling 110 gallons**, for outdoor use. Domestic well permits may limit outdoor watering — check ours.
- **Rabbit and chicken manure** = free fertilizer (compost chicken manure first; rabbit manure can go on directly).
- **Save seeds** from open-pollinated (heirloom) varieties. Keep a stash of seeds in a cool dry place — buy extra now.
- **Root cellar / basement storage:** potatoes, squash, carrots, onions, and cabbage keep for months in a cool, dark, slightly humid space.

Foraging around Wiggins — realistic expectations

The plains do **not** offer much wild food, especially in winter. Foraging is a **supplement**, not a plan. **Never eat a plant unless you are 100% sure what it is.** Don't feed foraged plants to the kids or Mom unless you are certain.

Common, useful wild foods here (learn them with a guide or CSU Extension class):

Plant	Where	Season	Notes
Dandelion	yards, fields	spring–fall	leaves, flowers, roots all edible
Lamb's quarters	gardens, disturbed ground	summer	like spinach; cook
Purslane	gardens, sidewalks	summer	eat raw or cooked
Curly dock	ditches, fields	spring	cook; contains oxalates — moderate amounts
Wild plum, chokecherry	river bottoms, fencerows	late summer	**chokecherry pits, leaves, and stems are poisonous** — eat only the pitted, cooked fruit (jelly, syrup)
Prickly pear cactus	dry grassland	summer–fall	pads and fruit; burn/scrape off the tiny spines (glochids)
Cattail	ponds, ditches, river	spring–summer	shoots and roots; only from **clean** water
Wild asparagus	irrigation ditches	spring	
Jerusalem artichoke (sunflower tubers)	ditches, field edges	fall–winter	tubers dug after frost

⚠ Deadly look-alikes near the South Platte

- **Water hemlock** — grows in wet places along rivers and ditches; **one of the most poisonous plants in North America.** Looks like wild carrot/parsnip. **Don't eat any wild plant in the carrot family.**
- **Poison hemlock** — common in Colorado ditches; purple-spotted stems; deadly.
- **Death camas** — onion-like bulbs with no onion smell.
- **Wild mushrooms** — don't, unless an expert identifies them.

Hunting and fishing

The area has **mule deer, white-tailed deer, pronghorn, pheasant, quail, doves, waterfowl, cottontails, and fish** (South Platte, Jackson Lake, area reservoirs).

- **Colorado Parks and Wildlife (CPW) licenses and seasons apply** — buy hunting and fishing licenses and learn the seasons **now**. Laws still apply in emergencies unless officials say otherwise. Hunt only where you have **landowner permission** or on public/walk-in access land.
- **Chronic Wasting Disease (CWD)** is present in Colorado deer and elk. CPW recommends **testing** animals and not eating meat from animals that test positive or look sick.
- Cook wild game thoroughly (**160°F** for ground meat).

- **Firearm safety with four kids:** guns **locked and unloaded**, ammunition stored separately, **keys/combination only known to adults.** Teach the 6-year-old: "Don't touch. Leave the room. Tell an adult."

Do it now

- [ ] Build the one-month pantry, then two, then three. Buckets, Mylar, oxygen absorbers.
- [ ] Hand grain mill + wheat berries; two manual can openers.
- [ ] Buy extra seeds; plan next year's garden (potatoes, squash, beans, corn).
- [ ] Two rain barrels (110 gal total).
- [ ] Hail cloth/row covers and a cold frame.
- [ ] CPW hunting & fishing licenses; learn a few local wild plants with an expert.
- [ ] Gun safe / lock boxes.

CHAPTER 8

Medical Basics: Cold, Carbon Monoxide & Emergencies

Plan for this: no 911, no ER, no pharmacy. Whatever happens, **we are the medical team.** This chapter covers the emergencies most likely to hit our family in a Colorado winter. **Chapter 9** covers wounds, infections, fevers, sickness, broken bones, and the medical supplies to stock.

> **If services ARE running:** call **911**. Nearest ER: **St. Elizabeth Hospital, 1000 Lincoln St, Fort Morgan, 970-867-3391** (ER entrance on Grant St). **Poison Control 1-800-222-1222.** Go early — don't wait for things to get worse.

The best medicine is training — get it now

Books help, but hands remember. **While classes still exist**, get:

- **CPR and First Aid, including infant and child CPR and choking** (Red Cross or American Heart Association) — Daniel, wife, Grandma, Grandpa.
- **Stop the Bleed** (free, about 1–2 hours) — tourniquets and wound packing.
- **Wilderness First Aid (WFA, 2 days)** or **Wilderness First Responder (WFR, ~9–10 days)** — from NOLS, SOLO, or Wilderness Medical Associates. These are built for **"help is hours or days away,"** which is exactly our plan.
- Ask our doctors for a **"long-term emergency" visit** (see Chapter 9): extra prescriptions, an emergency antibiotic plan, dosing for each child by weight.

Hypothermia (body too cold) *(CDC, Wilderness Medical Society)*

In a long outage, **hypothermia can happen indoors** — especially to the **newborn, toddlers, and Grandma and Grandpa**.

Signs — adults: shivering, clumsy hands, stumbling, slurred speech, confusion, drowsiness. **Shivering that STOPS while still cold is a bad sign. Infants:** bright red, cold skin and very low energy; weak cry; poor feeding.

What to do (no help coming):

1. Move them to the **warmest room**. Get them **off the floor** and out of wind.
2. **Remove wet clothes**; dry them.
3. **Wrap like a burrito:** sleeping bag or blankets, plus a **vapor barrier** (plastic sheet, trash bags, or emergency blanket) outside the insulation, and **cover the head and neck.**
4. **Heat to the chest, armpits, and back** — warm (not hot) water bottles wrapped in cloth, or **skin-to-skin** with a warm person inside the wrap. For a baby, **skin-to-skin on Mom or Dad's chest**, covered, is the best warmer there is.

5. **If awake and able to swallow:** warm, sweet drinks and food (sugar fuels shivering). No alcohol.
6. **Handle severe hypothermia gently** — rough movement can upset the heart. Keep them lying flat.
7. **Keep checking breathing.** Very cold people can have a slow, faint pulse — check for a full minute before deciding there is none.

Prevention is everything: one heated room, dry layers, hats, eat and drink regularly, check the babies and elders every hour at night.

Frostbite *(CDC, Wilderness Medical Society)*

Signs: skin turns red, then **white or grayish-yellow**, **hard or waxy**, and **numb**. Fingers, toes, ears, nose, cheeks.

What to do (no help coming):

1. **Treat hypothermia first.**
2. **Don't rewarm if it could freeze again** (e.g., you still have to walk home in the storm). Refreezing does far more damage. Walk on frozen feet if you must to reach shelter — then rewarm once safe.
3. **Rewarm fast in warm water — about 99–102°F** (comfortably warm to an elbow; use a thermometer if you have one) for **15–30 minutes**, until the skin is soft and color returns. Add warm water as it cools. It will **hurt a lot** — give ibuprofen (it also helps the tissue) and acetaminophen per Chapter 9 doses.
4. **Never rub with snow or hands, never use a stove, fire, or heating pad** — numb skin burns.
5. After thawing: **don't pop clear blisters**; put **aloe vera gel** on, wrap loosely with dry gauze **between fingers and toes**, elevate, keep it from refreezing.
6. **Blood-filled blisters or skin that stays black** = deep injury. Keep it clean and dry; dead tissue often separates on its own over weeks. Watch daily for **infection** (Chapter 9). **Don't try to cut away blackened tissue** — even surgeons wait.

Carbon monoxide poisoning *(CDC)*

The most likely way a grid-down winter kills a whole family. **Battery CO alarms in every sleeping area are not optional.**

Signs: headache, dizziness, weakness, nausea or vomiting, confusion, sleepiness. **Several people (or pets) sick at once indoors** = think CO first.

What to do: everyone out into fresh air immediately, open doors and windows, **shut off the source** (generator, heater, stove, blocked chimney). Keep the sick outside (bundled) or in a ventilated room until the alarm reads zero. Headache may last hours. **Fresh air is the only home treatment** — someone unconscious or confused after CO has a poor outlook without hospital oxygen, which is why **prevention is the real answer.**

Choking and CPR — what can still save a life

- **Choking — infants under 1:** 5 back blows (baby face-down on your forearm), then 5 chest thrusts with two fingers — repeat until it comes out or baby goes limp, then start infant CPR. **Children and adults:** abdominal thrusts (Heimlich).
- **CPR without hospitals — honest truth:** for an adult whose heart has stopped from a heart attack, CPR alone (no defibrillator or hospital) **rarely** brings them back. But **CPR with rescue breaths does save children and people who stopped breathing from choking, drowning, smoke, or CO** — because their heart is often still fine. **Learn it.**

Strokes and heart attacks (Grandma and Grandpa)

- **Heart attack signs:** chest pressure or pain spreading to arm, jaw, or back, sweating, short of breath, nausea. **Without services:** have them **rest sitting up**, keep them calm and warm, and if they are not allergic and not on blood thinners, **chew one regular (325 mg) aspirin** *(American Heart Association advice for suspected heart attack)*. Ask their doctor **now** whether that's right for them.
- **Stroke signs (BE FAST):** Balance, Eyes, Face drooping, Arm weakness, Speech trouble, Time. **Without services:** lay them on their side if drowsy, nothing by mouth until they can swallow safely, prevent falls, keep them warm and comfortable. Recovery care is mostly **nursing care at home** — feeding, turning, toileting.
- **Prevention now:** keep their blood pressure, heart, and diabetes medicines filled with **90-day supplies**, and get a **home blood-pressure cuff**.

Do it now

- [ ] CPR/First Aid with infant & child (all four adults); Stop the Bleed; a WFA or WFR course for Daniel and/or wife.
- [ ] Battery CO alarms in every sleeping area, both houses.
- [ ] Sleeping bags, wool blankets, emergency blankets, hand warmers.
- [ ] A cooking thermometer that reads water temperature (frostbite rewarming).
- [ ] Ask Grandma and Grandpa's doctor about 90-day supplies and the aspirin question.
- [ ] Build the home medical kit in Chapter 9.

CHAPTER 9

Home Medicine When No Help Is Coming

This chapter assumes the worst: no 911, no ER, no pharmacy, no doctor — for weeks or longer. It is written so that a careful parent can handle the injuries and illnesses a farm family is most likely to face, using what we stock **now**.

> **If services ARE running:** use them — **911**, **St. Elizabeth Hospital ER, 1000 Lincoln St, Fort Morgan (970-867-3391, entrance on Grant St)**, **Poison Control 1-800-222-1222**. Everything below is for when that isn't possible.

Part 1 — Mindset and triage

Be honest about what home care can and can't do

Most injuries and illnesses get better with good home care: cuts and scrapes, sprains, most simple broken bones, minor burns, colds, flu, stomach bugs, ear infections, most fevers in children over 3 months. **Good nursing — fluids, warmth, cleanliness, rest, food — is what saves most lives.** That was true for our great-grandparents, and it is true now.

Some things are very likely to be fatal without a hospital, and no book can change that: serious **internal bleeding** (e.g., from a crushed belly or a kicked chest), severe **head injuries** with worsening drowsiness, **appendicitis** that ruptures, **large deep burns** (more than about a quarter of the body), **sepsis** (blood infection) without antibiotics, **open breaks of the thigh**, severe **allergic reactions** without epinephrine, a **baby born very early** or a complicated birth, and a major **heart attack or stroke**. With these, we still do everything we can — **comfort, warmth, fluids, cleanliness, and love** — and some people do survive. Knowing this ahead of time helps us **prevent** these injuries (tractor, livestock, ladder, stove, and gun safety) and plan **honestly**.

The order of priorities (ABC + bleeding first)

1. **Danger:** make the scene safe (fire, CO, animals, machinery, ice).
2. **Massive bleeding:** stop it **first** — a person can bleed to death in minutes (Part 2).
3. **Airway:** can they breathe? Unconscious and breathing → roll onto their **side** (recovery position) so vomit drains.
4. **Breathing:** count breaths; look for blue lips, struggling.
5. **Circulation:** pulse, skin color and temperature; treat for shock — lay flat, keep warm.
6. **Everything else:** broken bones, burns, cleaning wounds, pain.

Set up a sick room

- One room, **away from the babies' sleeping area**, near the bucket toilet if possible.

- A bed you can reach from both sides, a bucket for vomit, a **trash bag** for dressings, a **handwashing station** (soap, water jug with a spout, basin), a **bleach spray bottle**.
- **One caregiver** for contagious illness when possible; that person washes hands and changes outer clothes before handling the babies.
- Good light (headlamp), a chair, a clock, a **notebook**.

Keep records — they are your "lab tests"

Without a doctor, **trends** tell you if someone is getting better or worse. Write every few hours:

Time	Temp	Pulse	Breaths/min	O_2 % (oximeter)	Alert?	Fluids in	Urine / diapers out	Meds given (dose)	Notes

Normal resting ranges *(standard pediatric and adult values; awake, calm)*:

Age	Pulse (beats/min)	Breaths/min
Newborn – 1 year	100–160	30–60
1–3 years	90–150	24–40
4–12 years	70–120	18–30
Adult	60–100	12–20

- **Fever raises pulse and breathing** — recheck an hour after fever medicine.
- **Pulse oximeter:** healthy people usually read **about 95–100%** (a little lower at our altitude is common). **Readings staying below about 92%** with fast breathing mean serious lung trouble.
- **Urine** is the best home test of fluids: an adult should pee **at least every 6–8 hours**, pale yellow.

Part 2 — Bleeding control *(Stop the Bleed, Wilderness Medical Society, military TCCC)*

Step 1: Direct pressure — hard and long

Put gauze or clean cloth on the wound and **press hard with both hands, straight down, for at least 10 minutes** (count it on a clock). Don't peek. If it soaks through, add more on top and press harder.

Step 2: Pack deep wounds

For a deep wound that keeps bleeding — especially in the **armpit, groin, or neck** where a tourniquet can't go:

1. Find where the blood is coming from.

2. **Stuff gauze into the wound** — all the way to the bottom — pushing it in with your fingers, filling the hole completely. **Hemostatic gauze** (QuikClot, Celox) works best; plain gauze or clean cloth works too.
3. **Press on top for at least 3–5 minutes** (hemostatic gauze needs at least 5).
4. Wrap tightly with a **pressure bandage** (an "Israeli" emergency bandage or an elastic wrap) to hold it.

Step 3: Tourniquet — arms and legs

For **life-threatening bleeding from an arm or leg** (spurting, pooling, soaking through), **use a tourniquet right away** — don't wait.

- Use a **real commercial tourniquet** (CAT or SOF-T from a trusted seller — **knock-offs break**). An improvised one needs a strap **at least 1½ inches wide** and a **windlass** (stick) to twist it tight — **a belt alone won't work.** *(WMS)*
- Place it **2–3 inches above the wound** (not on a joint), "high and tight" if unsure.
- **Tighten until the bleeding stops** and the pulse below it disappears. **It will hurt badly — that's normal.** A loose tourniquet makes bleeding worse.
- **Write the time** on it or on the skin.
- **Time limits (no hospital coming):** a tourniquet can stay on for **up to 2 hours** with little risk. **Within that time**, once the person is stable, pack and pressure-wrap the wound, then **slowly loosen** the tourniquet while watching. If bleeding stays controlled, leave it loose (in place, ready). If bleeding returns, re-tighten. **Don't loosen it "every so often" to let blood flow** — that's an old, harmful myth. *(WMS)* Honest truth: with no surgery available, a tourniquet that must stay on for many hours usually means losing the limb — **but it saves the life.**

Shock

After big blood loss: **pale, cold, sweaty, fast weak pulse, fast breathing, confused, thirsty.** Lay them flat, keep them **warm** (cold makes bleeding worse), raise legs if no leg injury, and if awake and not vomiting, give **small sips** of oral rehydration solution. Keep checking.

Part 3 — Wound care without a doctor

1. Clean — the step that prevents most infections

- **Wash your hands, wear gloves.**
- **Irrigate with at least 1 liter (about a quart) of drinkable water** — more for big or dirty wounds. **Clean drinking water works as well as sterile saline.** *(WMS)*
- **Pressure helps:** squirt with a **large syringe (20–60 mL)** — ideally with an 18–19-gauge irrigation tip or catheter — or poke a **pin-hole in a clean zip bag** and squeeze. Hold the wound open and flush from the inside out.
- **To make saline** (gentler, if you prefer): boil water 1 minute, then add **about 2 level teaspoons of table salt per liter (quart)** and let cool, covered. Use within 24 hours.

- **Animal bites** (rabies risk): scrub with **soap and water for 15 minutes**, then rinse with **povidone-iodine (Betadine) diluted** to the color of weak tea. *(WMS / WHO rabies guidance)*
- Don't keep pouring **hydrogen peroxide or rubbing alcohol** into wounds — they damage tissue.

2. Debride — remove dirt and dead tissue

- Pick out dirt, gravel, wood, and hair with **tweezers boiled for 10 minutes or wiped with alcohol**.
- Trim only **obviously dead, loose flaps** of skin or tissue with clean scissors. **Don't cut into healthy (bleeding, pink) tissue.**
- Ground-in dirt left in a wound causes infection and permanent "tattoo" marks — scrub gently with a soft brush and soapy water.

3. Close it — or leave it open?

This is the most important decision in home wound care. **Closing a dirty wound seals germs inside, and that can kill.**

OK to close (after thorough cleaning): **clean, straight cuts**, less than **about 12 hours old** (face up to 24), not from bites, not deep punctures, not crushed.

- **Wound-closure strips / butterfly strips:** best home choice. Dry the skin (tincture of benzoin helps them stick), lay strips across the cut **⅛ inch apart**, pulling edges together gently.
- **Skin glue** (medical tissue adhesive like Dermabond, or "liquid bandage"): small, clean, low-tension cuts. Hold edges together, paint glue **on top** — not inside the cut.
- **Hair apposition** (scalp cuts): twist a few hairs from each side together across the cut and dab with glue.

Leave OPEN: animal and human bites, punctures, dirty or crushed wounds, wounds older than ~12 hours, deep wounds you can't clean to the bottom, and any wound that looks infected. Pack lightly with **moist gauze**, cover, and change daily. These heal from the bottom up — slower, more scar, but **much safer.**

Delayed closure: a wound left open can sometimes be **closed with strips at day 3–5** if it is clean, pink, and has no sign of infection. *(WMS)*

Suturing (stitches) — the honest version

Stitching a wound **closes in any germs left behind**, and badly placed stitches can cause deep infection. **Most survival-medicine teachers say: if in doubt, don't suture.** Strips and glue handle most cuts; open healing handles the rest. If someone in the family wants this skill:

- Take a **class** (some WFR courses and survival-medicine workshops teach it) and **practice on pig's feet or pork belly** until it's easy.
- Only on **clean, fresh, simple** wounds, after excellent irrigation; never bites, punctures, or dirty wounds.

- Stitches come out in about **5 days on the face, 7–10 on the body, 10–14 over joints.**

4. Dress it and keep it clean

- A thin layer of **petroleum jelly, antibiotic ointment, or medical-grade honey**, then a **non-stick pad**, then gauze or a wrap.
- **Change once a day** (or if wet or dirty). Look, smell, and compare to yesterday — **write it in the log.**
- **Keep flies and dirt off** — cover fully, especially in summer around chickens, rabbits, and manure. Fly eggs (maggots) in a wound need flushing out.
- **Elevate** swollen arms and legs above the heart.
- **Tetanus:** get everyone's shots up to date **now** (adults every 10 years; a booster after 5 years for dirty wounds — CDC). With no services, there is no tetanus shot after an injury — **only prevention works.**

Part 4 — Infection when there are no antibiotics

Know the stages

Stage	What you see	What to do
Normal healing	a little pink at the edges, mild soreness, clear fluid the first 1–2 days	keep clean and covered
Local infection	spreading redness, swelling, warmth, worsening pain after day 2, pus	open it up, soaks, honey, elevate, rest (below)
Abscess	a tender, swollen, **soft, fluctuant ("squishy") lump** of pus	warm soaks; may need draining (below)
Spreading infection (cellulitis)	redness **growing** — mark its edge with a pen and check every few hours; **red streaks** up the limb	**serious** — all of the below, plus rest and fluids; antibiotics if you have a prescribed supply
Sepsis (blood infection)	fever OR low temperature, **fast pulse, fast breathing, confusion, mottled or gray skin, very little urine**, feels "like dying"	**life-threatening.** Fluids, warmth, rest, fever medicine. Without antibiotics, many people do not survive.

What we can do at home

- **Open and drain:** if a closed wound becomes infected, **remove the strips or stitches**, let it drain, irrigate, and leave it open.
- **Warm soaks or compresses** — 15–20 minutes, **4 times a day** — bring blood flow and help pus come to a head.
- **Honey dressings** (medical-grade) on infected open wounds; change daily.
- **Elevate and rest** the part; **splint** an infected hand or foot so it doesn't move.
- **Fluids, food, sleep.** The body does the real fighting.

- **Draining an abscess (only if it is clearly soft and pointing near the surface, not on the face, neck, hand, or near big blood vessels):** clean the skin, use a **sterile blade or needle** (boiled 10 minutes) to make **one small cut** at the softest, most-raised point, let the pus drain, gently press around (not on) it, irrigate, and **pack loosely** with a strip of moist gauze, changing daily. **Don't cut a hard, red lump** — keep up warm soaks until it softens. *(Where There Is No Doctor)*

Antibiotics — plan NOW, not later

- **Talk to our family doctor now** about a **prescribed emergency supply** of antibiotics for a family with young children far from care (some doctors and **telehealth services that make emergency kits** will do this). Ask which drugs, which **child-safe forms** (liquids need water and spoil — ask about chewables or powders), **doses by weight**, what each is for, and **allergies**. Write it all down and keep the paper with the medicine.
- **Fish and animal antibiotics are a real temptation — and a real danger.** They aren't made or inspected for people, the label may not match what's inside, and **using the wrong drug, wrong dose, or wrong length** can fail to cure the infection, breed resistant germs, or cause a dangerous **allergic reaction** with no hospital to treat it. **This book will not give doses for them.** Get a real prescription while you can.
- **Allergy check:** know who in the family is allergic to **penicillin or sulfa**. Stock **diphenhydramine** (Benadryl) — and ask the doctor about an **epinephrine auto-injector** if anyone has had a severe reaction.

Part 5 — Fever and sickness at home

Fluids are the #1 lifesaver

More children around the world die from **dehydration from diarrhea and vomiting** than from the germs themselves. **Oral rehydration solution (ORS) saves lives.** *(WHO)*

Store-bought ORS packets (WHO formula) or Pedialyte are best — stock plenty. If you run out:

> **Home ORS:** 1 liter (about 1 quart) of clean, boiled-and-cooled water + **6 level teaspoons sugar** + **½ level teaspoon salt.** Stir. Use within 24 hours. *(WHO / Rehydration Project)*

- **Measure exactly** — too much salt is dangerous for children; a little too dilute is harmless.
- **How much:** after each watery stool, under 2 years **¼–½ cup**; 2 and older **½–1 cup**; adults as much as they want. *(WHO)*
- **Vomiting:** give **1 teaspoon (5 mL) every 1–2 minutes** with a spoon or syringe. That small amount usually stays down and adds up to about a cup an hour. After 4 hours without vomiting, give more.

- **Babies: keep breastfeeding** — more often, not less. Breast milk is the best rehydration a baby can get. Formula-fed baby: keep formula at normal strength; offer ORS between feeds if the baby is dehydrated (WHO-type ORS packets are made for all ages; homemade ORS only if nothing else, in small amounts).
- **Older kids and adults: keep eating** bland foods (rice, crackers, bananas, soups, eggs) once they're hungry. Feeding during diarrhea helps recovery.

Judging dehydration in babies and kids *(WHO, AAP)*

	Some dehydration	Severe dehydration
Behavior	restless, irritable, **thirsty**	**very sleepy, floppy, hard to wake**
Eyes	sunken	very sunken; no tears
Mouth	dry	very dry
Soft spot (babies)	slightly sunken	clearly sunken
Wet diapers	fewer than usual	**none for 6–8+ hours**
Skin pinch (belly)	goes back slowly	goes back **very** slowly (2+ seconds)

Severe dehydration without IV fluids is deadly. Give ORS by spoon or syringe **constantly and patiently** — a teaspoon every minute, day and night if needed; keep breastfeeding. Many children can be saved this way.

Fever

Fever is the body fighting infection. In children over 3 months it is **usually not dangerous by itself** — how the child **acts** and **drinks** matters more than the number. *(AAP)*

- **Track it** on the log: temperature, time, medicine, drinking, behavior.
- **Fluids, light clothing, rest.** Lukewarm sponging if uncomfortable — **never cold water, ice, or rubbing alcohol.**
- **Medicine is for comfort** (doses below). It lets a child drink and sleep.
- **Warning signs (worse, not better):** hard to wake or very floppy; stiff neck; **purple/red rash that doesn't fade when you press a glass on it**; struggling to breathe; refusing all fluids; seizure; fever above 104°F that won't come down; fever over 3 days. With no help available: **all of the care in this section, constantly**, plus antibiotics if you have a prescribed plan for it.

A fever in our newborn (under 3 months)

In normal times, **any temperature of 100.4°F (38°C) or higher in a baby under 3 months means the ER immediately** — young babies can get blood, urine, or brain infections that look mild at first. *(AAP)* **With no help coming, here's what we can still do:**

1. **Recheck** the temperature (rectal is most accurate). Is the baby overdressed or next to the stove? Unbundle, recheck in 15–20 minutes.

2. **Keep breastfeeding — often.** Feeding is the baby's lifeline. If too weak to nurse, give **expressed breast milk by spoon, cup, or syringe** in small amounts often.
3. **Count wet diapers** and watch the soft spot.
4. **Keep the baby at a comfortable temperature** — not bundled hot, not chilled.
5. **Acetaminophen only if a doctor gave you a dose for this baby's weight in advance** (ask the pediatrician **now** what to do in this exact situation and write it here: _____).
6. **Watch for sepsis signs:** won't feed, very sleepy or limp, **grunting or fast breathing**, pale, mottled, or blue color, very low temperature (below 97°F), bulging soft spot, seizures.
7. **No ibuprofen, aspirin, honey, herbal teas, or water for a newborn.**

Honest truth: a very sick young baby can die without hospital care even with perfect home care. That's why **breastfeeding, handwashing, keeping sick people away from the baby, and vaccines on schedule now** matter so much.

Fever and pain medicine — doses by WEIGHT *(AAP / HealthyChildren.org dosing tables)*

Use weight, not age. Use the syringe that comes with the medicine, not a kitchen spoon. Check the strength on the box matches.

Acetaminophen (Tylenol) — liquid **160 mg / 5 mL**. Every **4–6 hours** as needed; **no more than 5 doses in 24 hours**.

Weight	Usual age	Dose
6–11 lb	0–3 months	**Only with a dose from the doctor**
12–17 lb	4–11 months	2.5 mL (80 mg)
18–23 lb	12–23 months	3.75 mL (120 mg)
24–35 lb	2–3 years	5 mL (160 mg)
36–47 lb	4–5 years	7.5 mL (240 mg)
48–59 lb	6–8 years	10 mL (320 mg)

Ibuprofen (Motrin, Advil) — **NOT under 6 months.** Every **6–8 hours** as needed; **no more than 4 doses in 24 hours**. Give with food; avoid when the child is dehydrated or vomiting a lot.

Weight	Usual age	Infant drops (50 mg / 1.25 mL)	Children's liquid (100 mg / 5 mL)
under 12 lb	under 6 months	**Do not use**	**Do not use**
12–17 lb	6–11 months	1.25 mL	2.5 mL
18–23 lb	12–23 months	1.875 mL	3.75 mL
24–35 lb	2–3 years	—	5 mL
36–47 lb	4–5 years	—	7.5 mL
48–59 lb	6–8 years	—	10 mL

- **Infant ibuprofen drops are twice as strong as children's liquid.**
- **NEVER aspirin, willow bark, or Pepto-Bismol (a salicylate) for children or teens** — Reye's syndrome. *(AAP, FDA)*
- **Don't double up** acetaminophen (many cold medicines contain it). **Write every dose down.**
- **Adults:** acetaminophen up to **3,000 mg/day**; ibuprofen **200–400 mg every 6–8 hours** (OTC label maximum 1,200 mg/day) with food. Breastfeeding Mom: both are generally considered compatible with nursing *(LactMed)*.

Vomiting and diarrhea — and stopping it from spreading

- ORS as above; **don't give anti-diarrhea medicine (loperamide/Imodium) to young children** (not under 2; label says 6 and up) — adults only, and **not with bloody diarrhea or high fever.**
- **Blood in stool with fever** = dysentery. Fluids, food, rest; this is one of the main reasons to have a prescribed antibiotic plan.
- **Stomach bugs (like norovirus) tear through a household.** Hand sanitizer **doesn't kill norovirus well — soap and running water does.** Clean vomit and diarrhea with **bleach solution: 5–25 tablespoons of household bleach per gallon of water** (stronger for heavy soiling) and let it sit **5 minutes**. *(CDC)* Wash soiled laundry hot; wear gloves.
- **The sick person doesn't cook.** Separate towels and cups. Keep them away from the babies.

Coughs, colds, and breathing trouble

- Most coughs and colds clear in 1–2 weeks. Rest, fluids, **humidified air** (a pot of water near — not on — the stove, or a steamy bathroom), **saline nose drops and a bulb syringe** for babies (they must breathe through their noses to nurse).
- **Honey** for cough in kids **over 1 year** (½–1 teaspoon) works about as well as cough syrup. *(AAP)* **No cough or cold medicines under 4.**
- **Croup** (barking, seal-like cough in toddlers, often at night): sit them up, keep calm, try **cool night air** (bundle up by an open door or window for a few minutes) or a steamy bathroom.
- **Danger signs (pneumonia or worse): fast breathing** (above the normal range table), **skin pulling in between ribs or at the neck**, nostrils flaring, **grunting**, blue or gray lips, oximeter **below ~92%**, too breathless to talk or feed. **With no help:** keep them **sitting up**, fluids in small sips, fever control, rest, warmth, oximeter checks, and antibiotics if prescribed for this. These are the illnesses where a home **pulse oximeter** earns its price.

Prevent the illness in the first place

- **Handwashing with soap** before food and after the toilet, diapers, and animals — the single most effective "medicine" in this book.
- **Safe water** (Chapter 2), **safe toilets** away from water and food, **flies kept off food.**
- **Quarantine new illness:** if neighbors are sick, limit contact for a few days.

- **Vaccines on schedule now** — whooping cough, measles, and flu kill babies when hospitals aren't there.

Part 6 — Broken bones, sprains, and dislocations

Sprains and strains

Rest, ice or cold (snow in a bag — 20 minutes on, wrapped in cloth), compression wrap, elevation. Ibuprofen helps. Start gentle movement after 2–3 days. A sprain that can't take **any** weight after a few days may be a break — splint it.

Broken bones (fractures)

Signs: deformity, severe pain at one spot, swelling, can't use it, grinding.

1. **Check circulation, feeling, and movement below the injury** (pulse, warm pink fingers or toes, can they feel your touch?).
2. **Splint it** so the joints **above and below** the break can't move. Use a **SAM splint**, boards, rolled magazines, or cardboard, padded well, tied with cloth or elastic wrap — **snug, not tight**. Fingers: tape to the next finger ("buddy taping"). Arm: sling and swathe to the body.
3. **Badly bent limb with no pulse beyond it:** gently pull **in line with the bone** (steady traction along its length) to bring it back to roughly straight, then splint. Stop if pain becomes extreme or you feel resistance. *(Wilderness First Responder teaching)*
4. Recheck circulation **every hour** — loosen if fingers/toes go cold, pale, numb, or blue.
5. **Healing time:** most simple breaks need **about 6 weeks** in a splint (less in young kids, more in older adults). Keep elevated the first days.

Open fracture (bone through skin): very dangerous without surgery and antibiotics. Control bleeding, **irrigate with lots of clean water**, cover with moist sterile gauze, splint, and watch for infection constantly. Use a prescribed antibiotic plan if you have one.

Thigh (femur) or pelvis fractures can bleed internally and are often fatal without a hospital. Keep the person **flat and still, warm**, splint both legs together, pain control, fluids.

Neck or back injury (fall from a hayloft, thrown from a horse, rollover): if neck pain, numbness, tingling, or weakness — keep the head and body **still and in line**, roll as a unit with helpers. Many neck injuries heal with rest and a padded collar (towel roll), but paralysis cannot be reversed at home.

Dislocations

- **Fingers** and **kneecaps** often go back with gentle, steady straightening (a kneecap usually slides back when the leg is slowly straightened). Splint afterward.
- **Shoulders** can sometimes be put back by trained people with slow, gentle techniques — **learn them in a WFR course**; don't yank. If it can't be put back, sling it in the most comfortable position.

- **Elbows, hips, and anything with a possible fracture: don't try** — splint as found.

Part 7 — Burns over the days that follow

1. **Cool running water for 20 minutes** — first, always. No ice, butter, or oil.
2. **Estimate size:** the person's **palm (with fingers) ≈ 1% of their body**. **Over about 10% in a child or 15% in an adult** loses enough fluid to be dangerous — **push ORS** hard (small amounts very often) and watch urine.
3. **Clean gently** each day with clean water and mild soap. **Leave blisters intact**; trim away blisters that have burst.
4. **Dress:** thin layer of **medical-grade honey** (good evidence for partial-thickness burns — Cochrane), **silver sulfadiazine** if prescribed, or petroleum jelly, then **non-stick pads** and loose gauze. **Change daily.** Separate burned fingers with gauze.
5. **Keep moving burned joints gently** a few times a day so they don't heal stiff; elevate swollen limbs.
6. **Pain:** ibuprofen and acetaminophen together (doses above).
7. **Watch for infection** (Part 4) — the main killer of burn patients after the first days.
8. **Deep burns** (white, leathery, painless, or charred) heal very slowly from the edges and scar; large ones are often fatal without a hospital. Keep clean, covered, fed, hydrated, and warm.

Prevention: stove gates, pot handles turned in, hot liquids out of kids' reach, no kids near the wood stove or burn barrel.

Part 8 — Toothaches and dental emergencies *(Where There Is No Dentist; American Dental Association)*

- **Pain:** for adults, the **ADA recommends ibuprofen plus acetaminophen together** as the best non-opioid dental pain relief. Kids: weight-based doses above.
- **Clove oil** (eugenol) on a cotton bit, held on the tooth, numbs pain (traditional, with some evidence). **Don't swallow it, and don't use on young children.**
- **Lost filling or broken tooth: temporary filling material** (pharmacy dental kits like Dentemp) covers it.
- **Warm salt-water rinses** (½ tsp salt in a cup of warm water) several times a day for sore gums and infections.
- **Knocked-out adult tooth:** hold by the crown (not the root), rinse briefly, **push back into the socket** within minutes if possible, bite on gauze; or store in milk. **Don't replant baby teeth.**
- **Danger:** a tooth infection with **swelling spreading to the eye, jaw, or neck, trouble swallowing or breathing, or high fever** can kill. Warm rinses, pain control, and a prescribed antibiotic plan; an abscess pointing in the gum can be drained like Part 4.
- **Prevention now:** dental checkups for everyone, fix problem teeth while dentists exist, **brush and floss** — and stock toothbrushes, floss, and fluoride toothpaste.

Part 9 — Herbs and kitchen remedies: what the evidence says

With pharmacies closed, people turn to plants. Some help with **comfort**; **none replace** fluids, wound cleaning, or antibiotics. **Strong** = good human studies; **Some** = a few studies or well-documented traditional use; **Weak** = mostly tradition or lab work.

Safety first:

- **Identify plants with 100% certainty** — learn them from CSU Extension or an experienced local now. Deadly look-alikes grow by the South Platte (water hemlock, poison hemlock — Chapter 7).
- **Breastfeeding Mom and the babies: avoid herbal medicines** beyond normal food amounts unless a doctor okays them; little is known about safety for infants. *(NCCIH)*
- **Ragweed/daisy family allergy:** avoid yarrow, chamomile, and echinacea.
- Herbs can interact with medicines (especially blood thinners).

Remedy	Where	Traditional use	Evidence	Cautions
Honey	kitchen / beekeepers	wounds, burns, cough	**Strong** (burns; cough in kids over 1)	**Never by mouth under 1 year**; medical-grade for wounds
Aloe vera gel	houseplant	minor burns, frostbite, sunburn	**Some**	skin only; patch test
Ginger (tea, fresh)	kitchen	nausea, stomach upset	**Some–Strong** for nausea	food amounts fine
Chamomile tea	garden	upset stomach, sleep, calm	**Weak–Some**	ragweed allergy; **not for babies**
Clove oil	kitchen	toothache	**Some**	don't swallow; not for young kids
Yarrow (*Achillea millefolium*)	native — roadsides, pastures	packed on small cuts to slow bleeding	**Weak** (traditional only) *(EMA)*	ragweed-family allergy; not in pregnancy; small superficial wounds only
Plantain (*Plantago major*)	yards, farmyards	crushed leaf on bites, stings	**Weak**	wash the wound first anyway
Prickly pear pad gel	dry pastures	burns, scrapes	**Weak**	remove every spine and glochid
Garlic	garden	eaten for colds	**Weak**	**NEVER raw on skin** — chemical burns, especially kids
Sage	garden	gargle for sore throat	**Weak–Some**	no concentrated oil; not large amounts in pregnancy
Elderberry (cooked syrup)	river bottoms	cold and flu symptoms	**Weak–Some** *(NCCIH)*	**raw berries, leaves, stems poisonous**; safety unknown for pregnancy, nursing, babies

Remedy	Where	Traditional use	Evidence	Cautions
Salt-water gargle	kitchen	sore throat, gums	**Some**	½ tsp salt per cup warm water
Willow bark	creeks	pain, fever	works like aspirin	**NEVER for children** (Reye's syndrome)

Part 10 — Dangers of our area, with no help coming

Rattlesnake bites

Prairie rattlesnakes live in the grasslands, rocky areas, and outbuildings of eastern Colorado, mostly April–October. *(Colorado Parks and Wildlife)*

The honest picture: antivenom (only in hospitals) is the real treatment. **Without it, most adults bitten by a prairie rattlesnake survive**, but bites can cause **severe swelling, tissue death, bleeding problems, and lasting damage**, and **small children are at much greater risk** because they get the same venom in a smaller body. Some bites are "dry" (little or no venom).

What to do:

1. **Move away from the snake.** Don't try to catch or kill it.
2. **Remove rings, watches, boots, tight clothes** from the bitten limb immediately.
3. Keep the person **calm, lying down, and still**; keep the limb **still (splint it) at about heart level.** *(Wilderness Medical Society)*
4. **Mark the edge of swelling** with a pen and the time, every 30 minutes.
5. **Fluids** by mouth if not vomiting; **acetaminophen for pain — NOT ibuprofen or aspirin** (they can worsen bleeding).
6. Watch for **shock** (Part 2) and **bleeding** from gums, nose, or old cuts; treat shock.
7. Over days: keep the limb elevated after the first hours, clean any blisters, and watch for infection.

Do NOT: cut the bite, **suck out venom** (no device works), use a **tourniquet or tight wrap**, put **ice** on it, give **alcohol**, or use electric shock. *(WMS, CDC)*

Prevention is the real treatment: boots and long pants, look before reaching into woodpiles, hay, and under coops and hutches, mow around buildings, teach kids to **freeze and back away**.

Frostbite and hypothermia

See Chapter 8 — rewarm frostbite in **warm water (about 99–102°F)** only when it won't refreeze; warm the core first for hypothermia.

Heat illness

- **Heat exhaustion** (heavy sweating, weak, dizzy, headache, nausea): shade, lie down, loosen clothes, wet cloths, sip ORS.

- **Heat stroke** (**hot skin, confusion, collapse**, may stop sweating) is deadly: **cool them as fast as possible** — soak with water and fan hard, or put them in a stock tank or tub of cool water (stay with them and keep the head up), ice or cold packs at neck, armpits, groin. Keep cooling until they're alert. *(CDC, WMS)*
- Babies, toddlers, and Grandma and Grandpa overheat fastest.

Dehydration — dry air and altitude

Our air is **very dry** and we're at **about 4,500 feet** — you lose water just breathing, working outside, or sitting by the wood stove. **Drink before you're thirsty**; check that urine is pale. Nursing Mom needs extra.

Hantavirus — from mouse dust

A rare but **often deadly** lung infection from **deer mice** in barns, sheds, coops, feed rooms, and vehicles. Caught by **breathing dust** from droppings, urine, and nests. Colorado has cases most years. **There is no home cure — prevention is everything.** *(CDC, CDPHE)*

Safe cleanup (CDC): open doors and windows **30 minutes** first and leave; **gloves on**; **do NOT sweep, vacuum, or leaf-blow**; spray droppings and nests **until soaking wet** with **1½ cups bleach per gallon of water** (fresh); **soak 5 minutes**; wipe up with paper towels, bag, mop with bleach solution; wash gloves, then hands. A well-fitted **N95 mask** adds protection.

Symptoms (1–8 weeks later): fever, aching thighs, hips, and back, fatigue, then **cough and shortness of breath.** With no hospital: rest, fluids, fever control, sitting up, oxygen checks — and honestly, severe cases are often fatal even in hospitals.

Prevention: feed in **metal cans with lids**; seal gaps; trap mice; keep spilled feed cleaned up.

Tularemia — from rabbits

"Rabbit fever," a bacterial infection from **rabbits and rodents**, **tick and deer fly bites**, or **dust** (mowing over a dead rabbit). Seen in Colorado most years. *(CDC, CDPHE)*

- **Wear gloves when butchering** (Chapter 6); don't butcher a rabbit that seemed sick; **never handle wild rabbits found dead** bare-handed. **Cook rabbit to 160°F.**
- Repellent and tick checks.
- **Symptoms:** sudden high fever, chills, swollen lymph nodes, a sore where it entered. It is **treatable with the right antibiotics** — another reason for a prescribed plan. Without them: rest, fluids, fever care.

Also in our area

- **West Nile virus** (mosquitoes by the river and ditches): repellent at dawn and dusk, dump standing water. Most cases are mild; older adults are at highest risk.
- **Plague** (fleas on **prairie dogs** and rodents): keep pets away from colonies, flea control, don't handle dead rodents.
- **Salmonella** from chickens: wash hands after birds and eggs (Chapter 5).

- **Rabies** (skunks, bats): **there is no treatment after symptoms start, and no shots without services.** Keep dogs and cats vaccinated, keep kids away from wild animals, and scrub any bite for 15 minutes.

Part 11 — Stock up NOW: the home medical kit and pharmacy

*Rough quantities for **8 people for about 3 months**. Store cool, dry, and dark, **locked away from kids**. Rotate by date — use the oldest in normal life and replace.*

About expiration dates: the U.S. military's Shelf Life Extension Program found many **tablets and capsules** stay potent well past their dates when stored well; **liquids, children's syrups, eye drops, epinephrine, and insulin do not** — replace those on time. Never use medicine that has changed color, smell, or texture.

Fever, pain, and sickness (with kids' forms)

- ☐ Adult acetaminophen 500 mg — 2 bottles of 250+
- ☐ Adult ibuprofen 200 mg — 2 bottles of 250+
- ☐ Children's acetaminophen liquid (160 mg/5 mL) — 4–6 bottles; chewables for the 6-year-old
- ☐ Infants' ibuprofen drops — 2–3; children's ibuprofen liquid — 4–6
- ☐ Oral dosing syringes — 6+ (they get lost)
- ☐ **ORS packets (WHO formula) — 100+**, plus Pedialyte; extra sugar and salt with the recipe taped on
- ☐ Diphenhydramine (adult tablets + children's liquid) and cetirizine or loratadine (adult + children's)
- ☐ Hydrocortisone 1% cream; antifungal cream (clotrimazole); diaper cream (zinc)
- ☐ Antacids / famotidine; loperamide (adults only); fiber or stool softener
- ☐ Saline nose spray/drops + bulb syringe or nasal aspirator; honey
- ☐ Oral thrush/yeast and pinworm treatments (ask pharmacist)
- ☐ Multivitamins (adult + children's), prenatal vitamins for Mom
- ☐ **Prescriptions: 90-day supplies** for everyone (especially Grandma and Grandpa); **epinephrine** if anyone has severe allergies; a **doctor-prescribed antibiotic plan** with written doses

Bleeding and wounds

- ☐ **Tourniquets (CAT or SOF-T, genuine) — 3** (house, truck, barn); learn them
- ☐ Hemostatic gauze (QuikClot / Celox) — 4
- ☐ Emergency "Israeli" pressure bandages — 4; elastic wraps (2–6") — 8
- ☐ Gauze pads 4×4 — 300+; rolled gauze — 30; non-stick pads — 100; ABD/trauma pads — 20
- ☐ Medical tape (several types) — 10 rolls; adhesive bandages — 500 assorted
- ☐ **Wound-closure strips — 20 packs**; tincture of benzoin; skin glue / liquid bandage
- ☐ **Irrigation syringes (35–60 mL) with tips — 6**; povidone-iodine — 2 bottles

- [ ] **Medical-grade honey** (wound gel) — 4 tubes; petroleum jelly — 2 large; antibiotic ointment — 6
- [ ] Burn dressings / gel; aloe vera gel
- [ ] **Nitrile gloves — 3–4 boxes**; trauma shears, fine tweezers, splinter forceps, magnifier
- [ ] Suture practice kit and skin stapler **only if trained**; scalpel blades (for abscesses, trained use)

Bones, joints, and moving people

- [ ] **SAM splints — 3–4**; triangular bandages (slings) — 6; finger splints
- [ ] Instant cold packs — 10; reusable hot/cold packs
- [ ] Crutches (one pair, adjustable); a sturdy blanket or tarp to carry someone

Checking and nursing

- [ ] **Digital thermometers — 3** (one **rectal** for the baby) + batteries
- [ ] **Pulse oximeters — 2** (adult and pediatric-capable) + batteries
- [ ] **Blood-pressure cuff** (automatic) for Grandma and Grandpa; stethoscope; penlight
- [ ] **N95 masks — 50+**; surgical masks — 100; safety glasses
- [ ] Bedpan, urinal, waterproof pads, basins, plenty of soap, **bleach**, spray bottles, trash bags
- [ ] Notebook and pens for the **vitals log**; this book

Teeth and eyes

- [ ] Temporary filling kit (Dentemp) — 2; clove oil; dental floss; extra toothbrushes and toothpaste
- [ ] Eye wash (sterile saline) — 4; eye patches; spare glasses and contacts
- [ ] Reading glasses; hearing-aid batteries

Books and training (now)

- [ ] *Where There Is No Doctor* and *Where There Is No Dentist* (Hesperian — free PDFs; print them)
- [ ] *The Survival Medicine Handbook* (Joe Alton, MD & Amy Alton, APRN)
- [ ] A wilderness first aid manual from your WFA/WFR course (e.g., NOLS)
- [ ] Classes: **Stop the Bleed, CPR + pediatric CPR, WFA or WFR**

CHAPTER 10

Power, Light & Communication

Power

Generators — safe use *(CDC, FEMA)*

- **Outdoors only, at least 20 feet from windows, doors, and vents.** Never in the house, basement, garage, or barn with the door open.
- **Never backfeed** through a house outlet. Use a **transfer switch or interlock** installed by a licensed electrician, or plug appliances directly into the generator with heavy outdoor extension cords.
- Let it cool before refueling. Store fuel in approved cans, away from the house; add stabilizer and rotate.
- **Priority loads:** furnace blower/controls, well pump, freezer/fridge (run a few hours at a time), lights, phone charging, coop waterer. Check each one's watts.
- An **inverter generator** is quieter and safer for electronics.

Without a generator

- **Portable power stations** (battery boxes) + **solar panels** charge phones, lights, radios, a breast pump, and small fans.
- **Car** = phone charger (run it outside, never in a closed garage; clear snow from the tailpipe).
- **Batteries:** AA/AAA stock for radios and lights; rechargeables + solar charger.

Light

- **LED headlamps** for every adult (hands-free for diapers and chores) and LED lanterns for rooms.
- **Avoid candles and oil lamps** around toddlers — a leading cause of house fires in outages. If used, never leave burning unattended.
- Glow sticks for kids' nightlights.

Communication and information

Getting warnings

- **NOAA Weather Radio** with battery and hand-crank backup and **SAME** alerts set for **Morgan County**. Our forecasts come from the **National Weather Service office in Boulder** (Denver/Boulder). Write our local transmitter frequency in the Home Worksheet.

- **AM radio: KOA 850 AM (Denver)** is Colorado's main emergency broadcast station and carries far at night.
- Phones: turn on **Wireless Emergency Alerts**; sign up for Morgan County alerts.

Talking to each other

Option	License?	Use for
FRS walkie-talkies	No license	Between our house and Grandma & Grandpa's, the barn, the coop. Short range.
GMRS radios	**FCC license — no test**, one license covers the family	Longer range than FRS, can use repeaters.
Amateur (ham) radio	**Exam required** (Technician license is the entry level)	Talk across the county or state; local ham clubs often help in disasters.
CB radio	No license	Truckers and farm neighbors; short-medium range.

- Agree on **check-in times** (e.g., on the hour) and a **family channel**.
- **Out-of-area contact:** long-distance texts often work when local calls don't.
- **Texts beat calls** when networks are jammed.

Do it now

- [ ] NOAA Weather Radio (battery + crank), set to Morgan County.
- [ ] Headlamps for adults, lanterns, batteries.
- [ ] FRS/GMRS radio set (get the GMRS license online from the FCC).
- [ ] Power station + solar panel, or a generator with a professionally installed inlet.
- [ ] Phone car chargers and power banks.

CHAPTER 11

Security, Neighbors & Working Together

Families survive hard times **together**. Having Grandma and Grandpa next door is a huge strength. Good neighbors are worth more than any supply.

Two houses, one team

- **Decide now:** if heat or water runs short, **which house do we consolidate into?** (Usually the one with the reconnected wood stove, water access, and room for babies.)
- **Divide the jobs:**

Job	Who (fill in)
Babies & kids	
Heat & firewood	
Water	
Chickens & rabbits	
Cooking & food inventory	
Radio, news & check-ins	
Night watch / fire & CO watch	
Medical & medicines	

- **Daily family meeting** (5 minutes): what we have, what we need, today's jobs, health check on everyone.
- Grandma and Grandpa's knowledge (cooking from scratch, gardening, preserving) is valuable — and their health needs (medicines, mobility, staying warm) come first too.

Neighbors

- **Know them before trouble.** Trade phone numbers; learn who has skills (nurse, mechanic, electrician, farmer, hunter) and needs (elderly, medical equipment, new babies).
- **Trade and share:** eggs, rabbit, kits, hay, firewood, and labor are excellent trade goods. A neighbor network can watch for fire, share a generator, or butcher together.
- Join or start a simple **neighborhood check-in** (phone tree or radio channel).

Staying safe

- **Keep a low profile.** Don't advertise how much food, fuel, or water we have. Stack supplies out of sight.

- **Lock up:** house, outbuildings, fuel, feed, and generator (chain it). Animal theft rises in hard times.
- **Motion lights** (solar) and a **dog** are good early warnings.
- **Night watch** in rotation during unrest — also watching for chimney fires and CO alarms.
- **Avoid confrontation.** Most problems are solved by staying home, staying calm, and calling the Sheriff.
- **Firearms:** only adults trained and lawful; **locked storage** with four kids in the home — always. Know Colorado law on self-defense and safe storage.
- **Scams and rumors:** check news from official sources (Sheriff, NWS, county) before acting.

Taking care of our heads

- Stress is normal. Keep routines, sleep in shifts, eat regularly, step outside for daylight.
- Talk to each other. Give the 6-year-old honest, simple answers.
- Watch for signs a family member is struggling (not sleeping or eating, withdrawing, hopelessness). Colorado Crisis Services and the **988** Suicide & Crisis Lifeline help by phone or text.

Do it now

- [ ] Sit down with Grandma and Grandpa: consolidation plan, job list, check-in plan.
- [ ] Neighbor phone list (Home Worksheet).
- [ ] Locks, solar motion lights, gun safe.

CHAPTER 12

Plains Weather & Disasters

Know the threats where we live and what to do in each.

Blizzards and extreme cold

- **Before:** fill propane, stock firewood, food, water, medicines; charge everything; bring animals' water and feed in close.
- **During: stay inside.** Plains blizzards cause **whiteouts** — people get lost between the house and barn. If you must go to the animals, **tie a rope line** from house to barn and go in pairs.
- **Never travel** in a blizzard warning unless it's a life-or-death emergency.
- **Car winter kit:** blankets/sleeping bags, hats, gloves, boots, water, snacks, shovel, ice scraper, sand/kitty litter, jumper cables, flashlight, phone charger, flares, and baby supplies when kids ride.
- **If stranded in a car:** stay with the car, run the engine ~10 minutes an hour for heat with the **tailpipe cleared of snow** and a window cracked, tie bright cloth to the antenna. *(NWS / Ready.gov)*

Lessons from the December 2006 Holiday Blizzards

- **Two blizzards hit eastern Colorado about a week apart** (Dec 20–22 and Dec 28–30). Drifts reached **10–15 feet**; roads closed and homes were cut off; the National Guard and Civil Air Patrol searched for stranded drivers and dropped hay. *(Colorado DHSEM after-action reports)*
- **Thousands of cattle died** (estimates of 8,000–15,000). Many ranchers **had hay in the field but couldn't reach it** through the drifts. In a 1997 storm, about 27,000–30,000 cattle died.
- **Lessons for us:** expect a **second storm**; don't use up supplies in the first. **Stage feed, hay, and water right next to the animals** before the storm, not across a field. Keep a shovel at every door, and a sled or wagon to haul feed and water.

Tornadoes

- **Watch** = conditions are right. **Warning** = take shelter **now**.
- Go to the **basement** (or an interior room on the lowest floor without windows). Get under something sturdy; cover heads — **bike helmets for kids** help.
- Keep **shoes, a flashlight, radio, and baby bag** in the shelter area.
- Vehicles and mobile homes are not safe.

Hail and lightning

- Hail can be big here — stay inside away from windows; protect gardens with hail cloth; park under cover.
- **"When thunder roars, go indoors."** Wait 30 minutes after the last thunder. *(NWS)*

Grass and wildfire

- On **Red Flag Warning** days (dry + windy), no outdoor burning, welding, or driving through dry grass.
- **Defensible space:** mow and clear dry grass and weeds around the house, coop, hutches, propane tank, and woodpile.
- **Evacuate early** if officials say go — wind-driven grass fires move very fast. Know two ways out.
- Plan how to move or release animals.

Flooding (South Platte and ditches)

- **Turn around, don't drown. 6 inches of moving water can knock an adult down; 12 inches can carry away most cars.** *(NWS)*
- Move to higher ground if water rises. Keep kids away from floodwater (sewage, chemicals).
- After flooding, **treat well water as contaminated** until tested (Chapter 2).

Extreme heat and drought

- See "Summer heat" in Chapter 3. Water for people and animals first. Shade for rabbits and chickens (rabbits die from heat quickly).

Do it now

- [ ] Car winter kits in every vehicle.
- [ ] Shelter spot in the basement stocked (helmets, shoes, light, radio, baby bag).
- [ ] Mow defensible space around buildings and propane tank.
- [ ] Rope for a house-to-barn line.

CHAPTER 13

Seasonal Checklists & Inventory Sheets

Print these pages extra. Use pencil.

Fall (September–October) — the most important season

- [ ] **Propane tank filled**; gauge recorded.
- [ ] **Chimney swept/inspected**; stove tested with CO alarms on.
- [ ] Firewood stacked, covered, dry.
- [ ] CO and smoke alarms tested; new batteries.
- [ ] Hay for the whole winter; layer feed and pellets stocked.
- [ ] Heated waterers working; backup crocks/buckets ready.
- [ ] Coop light on timer; coop and hutches wind-blocked.
- [ ] Winter car kits in vehicles.
- [ ] Rotate water storage; check formula, diaper sizes, medicines, baby clothes sizes.
- [ ] Squash, pumpkins, potatoes stored.
- [ ] Generator test run (outdoors) with fresh fuel.
- [ ] Review this book with Grandma and Grandpa.

Winter (November–March)

- [ ] Watch the forecast daily; fill water before storms.
- [ ] Record propane % weekly.
- [ ] Keep snow off the propane tank lid/regulator and away from furnace vents outside.
- [ ] Mealworm bin going; sprouting fodder for hens.
- [ ] Plan breeding so kits aren't born in deep cold.
- [ ] Order garden seeds.

Spring (April–May)

- [ ] Order/hatch chicks to replace old hens.
- [ ] Start seeds indoors; plant cool crops; potatoes after soil warms.
- [ ] Check stored food; rotate.
- [ ] Tornado shelter spot restocked.
- [ ] Rotate water storage (every 6 months).

Summer (June–August)

- [ ] Garden: water, mulch, hail protection.
- [ ] Black soldier fly bin running; dry extra larvae for winter.
- [ ] Shade and cool water for rabbits and hens.
- [ ] Mow defensible space (fire season).
- [ ] Buy firewood early (cheaper, drier).
- [ ] Start the mealworm colony for winter.

Go-bag (one per person, by the door)

- [ ] Water (1 gal/person/day for 3 days if carrying is possible) + filter
- [ ] 3 days of food that needs no cooking; kids' snacks
- [ ] Baby: diapers, wipes, ready-to-feed formula + nipples (for the 1-year-old if still on formula; small backup for the newborn), sleep sack, change of clothes, carrier/wrap
- [ ] Medicines + list; copies of IDs, insurance, birth certificates (in a zip bag); cash
- [ ] Warm clothes, hats, gloves, rain gear, sturdy shoes
- [ ] Flashlight/headlamp, batteries, radio, phone charger/power bank
- [ ] First aid kit, hand sanitizer, toilet paper
- [ ] This book; a paper map of the county; a whistle
- [ ] Comfort item for each kid

Inventory sheet — supplies

Item	Target	On hand	Date checked	Expires
Drinking water (gal)	125			
Unscented bleach	1 gal			
Infant formula — ready-to-feed (backup)				
1-yr-old formula/milk (oz or cans)				
UHT/powdered milk				
Diapers — size ___				
Diapers — size ___				
Wipes				
Prenatal vitamins				
Grains (lb)	150/month			
Beans (lb)	40/month			
Oil/lard (gal)	2/month			
Canned goods (cans)				
Salt / sugar (lb)				
Layer feed (lb)				
Oyster shell (lb)				
Rabbit pellets (lb)				
Hay (bales)				
Propane (%)	> 30%			
Firewood (cords)				
Generator fuel (gal)				
Batteries AA / AAA / D				
Prescription meds (days)	30+			
First aid kit checked	yes			

Daily log (during an emergency)

Date	Propane %	Firewood used	Water used (gal)	Eggs	Rabbits	Notes (health, weather, events)

CHAPTER 14

Sources & Where to Learn More

The figures in this book come from these organizations. Visit them (while the internet works) for the latest guidance, and print what's useful.

Emergency planning

- FEMA / **Ready.gov** — water storage (1 gallon per person per day; 2-week home supply), winter storms, tornadoes, floods, generator safety, go-bags. ready.gov
- **National Weather Service** (Denver/Boulder office) — forecasts, NOAA Weather Radio, winter car safety, flood ("Turn Around, Don't Drown") and lightning safety. weather.gov/bou
- **Morgan County** Office of Emergency Management / Sheriff — local alerts and shelters.

Water

- **CDC** — making water safe in an emergency (boiling 1 minute; bleach use). cdc.gov/healthywater
- **EPA** — emergency disinfection of drinking water (bleach drop chart). epa.gov
- **Colorado Division of Water Resources** — well permits and records; rain barrel law (2 barrels / 110 gallons).

Heat and fire safety

- **NFPA 211** — Standard for Chimneys, Fireplaces, Vents, and Solid Fuel-Burning Appliances (clearances: 36 in for unlisted stoves, 18 in for single-wall connector, unless listed/protected). nfpa.org
- **EPA Burn Wise** — safe wood burning, seasoned firewood (<20% moisture), EPA-certified stoves. epa.gov/burnwise
- **Chimney Safety Institute of America (CSIA)** — certified sweeps, inspection levels, liners. csia.org
- **CDC** — carbon monoxide poisoning prevention; hypothermia and frostbite. cdc.gov
- Propane energy content (about 91,500 BTU/gallon) — U.S. Energy Information Administration / propane industry standard.

Babies and kids

- **American Academy of Pediatrics (AAP) / HealthyChildren.org** — breastfeeding frequency, switching to whole milk at 12 months (16–24 oz/day), never dilute or make homemade formula, safe sleep, fever under 3 months, dressing infants.
- **CDC** — breastfeeding mothers' extra calories (~450–500/day), breast milk storage, infant formula preparation and safe water, Salmonella from backyard poultry.
- **FDA** — warnings against homemade infant formula.
- **LactMed** (National Library of Medicine) — medicines during breastfeeding.

Food, chickens, and rabbits

- **Colorado State University (CSU) Extension** — backyard poultry, rabbits, gardening on the plains, frost dates, food preservation, wild plants. extension.colostate.edu
- **USDA** — food safety during power outages (freezer 48/24 hours, fridge 4 hours), safe cooking temperatures (rabbit and game 160°F), nutrient data (eggs, rabbit), Dietary Guidelines calorie estimates. usda.gov, fooddata central

- **National Center for Home Food Preservation** — tested pressure-canning recipes for meat. nchfp.uga.edu
- **Colorado Department of Agriculture** — State Veterinarian, avian influenza reporting.

Hunting, fishing, foraging

- **Colorado Parks and Wildlife** — licenses, seasons, CWD testing. cpw.state.co.us

Medicine when no help is coming

- **Wilderness Medical Society (WMS) Practice Guidelines** — basic wound management in the austere environment (irrigate with at least 1 L of drinkable water; tap water as good as sterile saline; tourniquets ≥1.5 in wide with a windlass, up to 2 hours with minimal risk, no "intermittent release"; closure timing and delayed closure); frostbite, hypothermia, heat illness, and snakebite guidelines. wms.org
- **NOLS Wilderness Medicine**, **SOLO**, and **Wilderness Medical Associates** — Wilderness First Aid / First Responder training (splinting, traction-in-line for angulated fractures, simple dislocations, long-term patient care, vitals logs).
- **Tactical Combat Casualty Care (TCCC)** / **Stop the Bleed** — tourniquets, wound packing, hemostatic gauze, pressure bandages.
- ***Where There Is No Doctor*** (David Werner) and ***Where There Is No Dentist*** (Murray Dickson), **Hesperian Health Guides** — free online; wound care, leaving dirty wounds open, draining abscesses, dehydration, dental emergencies. hesperian.org
- ***The Survival Medicine Handbook*** (Joseph Alton, MD, and Amy Alton, APRN) — grid-down family medicine, medical storage, honest limits of home care, cautions on fish antibiotics.
- **World Health Organization (WHO)** — oral rehydration therapy, dehydration assessment, rabies wound washing (15 minutes soap and water); **Rehydration Project** (rehydrate.org) home ORS recipe (1 L water, 6 level tsp sugar, ½ level tsp salt).
- **AAP / HealthyChildren.org** — acetaminophen and ibuprofen dosing tables by weight (checked against Northwell Cohen Children's and St. Louis Children's Hospital charts), fever in infants, honey for cough over age 1, no cough/cold medicines for young children.
- **American Red Cross**, **American Heart Association** — CPR, choking, first aid, aspirin for suspected heart attack. **American Burn Association** — cool burns 20 minutes. **Cochrane Review** (Jull et al.) — honey for partial-thickness burns.
- **CDC** — tetanus boosters, hantavirus cleanup (ventilate 30 min, no sweeping, 1½ cups bleach/gallon, soak 5 min), norovirus cleanup (5–25 tbsp bleach/gallon), tularemia, plague, West Nile, rabies, heat illness, snakebite first aid, CO poisoning, frostbite and hypothermia. **CDPHE** — Colorado hantavirus, tularemia, plague, West Nile, rabies. **Colorado Parks and Wildlife** — prairie rattlesnakes.
- **American Dental Association** — ibuprofen plus acetaminophen for dental pain.
- **U.S. DoD/FDA Shelf Life Extension Program (SLEP)** — many solid medicines stay potent past their dates when stored well.
- **NCCIH** (elderberry; herbs in pregnancy/breastfeeding), **European Medicines Agency** (yarrow monograph), **LactMed** (medicines while breastfeeding), **FDA** (no aspirin for children, cough/cold medicines in young children).
- **CommonSpirit St. Elizabeth Hospital, Fort Morgan** — ER location and phone, for when services are running (hospital website, October 2026).
- **CSU Extension** — identifying local plants and poisonous look-alikes.

Garage and stored-water heating

- **CDC** — carbon monoxide: never run vehicles, generators, grills, or camp stoves in a garage, even with the door open.
- **NFPA** — space heater safety (3-foot clearance), heating-equipment fires, flammable liquids in garages; **NFPA 211** for any solid-fuel stove.
- **Indoor-rated portable propane heater manufacturers' manuals** (e.g., Mr. Heater "Buddy") — ODS and tip-over shutoffs, required ventilation, approved cylinders and hoses. **U.S. Consumer Product Safety Commission (CPSC)** — portable heater and CO alarm guidance.
- **CSU Extension** and other university Extension services — root-cellar practice of using water's stored heat to prevent freezing; hot composting temperatures and pile size.
- Basic physics of water (expands about 9% when frozen; releases heat as it freezes) — standard references.

Experienced preppers, homesteaders, and survival guides

- **U.S. Army Survival Manual FM 21-76** (now **ATP 3-50.21**) — the "SURVIVAL" mindset and priorities.
- **The Church of Jesus Christ of Latter-day Saints** — "All Is Safely Gathered In," Longer-Term Food Supply, and the one-year basics list (400 lb grains, 60 lb legumes, etc. per adult; child percentages); **BYU** Department of Nutrition, Dietetics & Food Science shelf-life research. churchofjesuschrist.org/topics/food-storage
- **The Prepared** (theprepared.com) — winter survival and lessons from the 2021 Texas blackout (tent-in-a-room, body heat, diesel gelling, indoor-rated heaters only).
- **Joel Salatin / Polyface Farm** — the Raken (rabbit-over-chicken) house and deep-bedding method (8–12+ inches of carbon). Mother Earth News; McMurray Hatchery blog.
- **Colorado Division of Homeland Security & Emergency Management** — after-action reports on the December 2006 Holiday Blizzards; **Natural Hazards Center (CU Boulder)** Quick Response Report 191 on livestock losses.
- **Recommended books for the shelf** (classic, widely respected references): *SAS Survival Handbook* (John "Lofty" Wiseman); *When Technology Fails* (Matthew Stein) — off-grid water, heat, and food; *Storey's Guide to Raising Chickens* (Gail Damerow); *Storey's Guide to Raising Rabbits* (Bob Bennett); *The Encyclopedia of Country Living* (Carla Emery).

Medical training

- **American Red Cross** and **American Heart Association** — First Aid, CPR (adult/child/infant). **Stop the Bleed** — bleeding control. 988 Suicide & Crisis Lifeline. Poison Control 1-800-222-1222.

This handbook was written for our family in October 2026. It is general guidance and not a substitute for professional medical, electrical, chimney, or legal advice. Check guidance yearly — recommendations change.